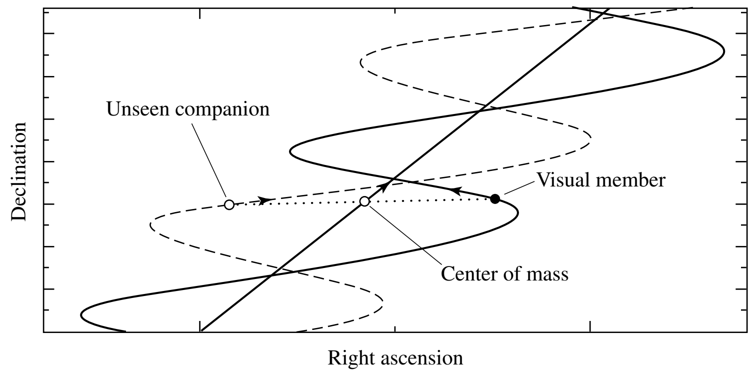

FIGURE 7.1 An astrometric binary, which contains one visible member. The unseen component is implied by the oscillatory motion of the observable star in the system. The proper motion of the entire system is reflected in the straight-line motion of the center of mass.

- **Astrometric binary**. If one member of a binary is significantly brighter than the other, it may not be possible to observe both members directly. In such a case the existence of the unseen member may be deduced by observing the oscillatory motion of the visible component. Since Newton's first law requires that a constant velocity be maintained by a mass unless a force is acting upon it, such an oscillatory behavior requires that another mass be present (see Fig. 7.1).

- **Eclipsing binary**. For binaries that have orbital planes oriented approximately along the line of sight of the observer, one star may periodically pass in front of the other, blocking the light of the eclipsed component (see Fig. 7.2). Such a system is recognizable by regular variations in the amount of light received at the telescope. Not only do observations of these *light curves* reveal the presence of two stars, but the data can also provide information about relative effective temperatures and radii of each star based on the depths of the light curve minima and the lengths of the eclipses. Details of such an analysis will be discussed in Section 7.3.

- **Spectrum binary**. A spectrum binary is a system with two superimposed, independent, discernible spectra. The Doppler effect (Eq. 4.35) causes the spectral lines of a star to be shifted from their rest frame wavelengths if that star has a nonzero radial velocity. Since the stars in a binary system are constantly in motion about their mutual center of mass, there must necessarily be periodic shifts in the wavelength of every spectral line of each star (unless the orbital plane is exactly perpendicular to the line of sight, of course). It is also apparent that when the lines of one star are blueshifted, the lines of the other must be redshifted relative to the wavelengths that would be produced if the stars were moving with the constant velocity of the center of mass. However, it may be that the orbital period is so long that the time dependence of the spectral wavelengths is not readily apparent. In any case, if one star is not overwhelmingly more luminous than its companion and if it is not possible to resolve each star separately, it may still be possible to recognize the object as a binary system by observing the superimposed and oppositely Doppler-shifted spectra.

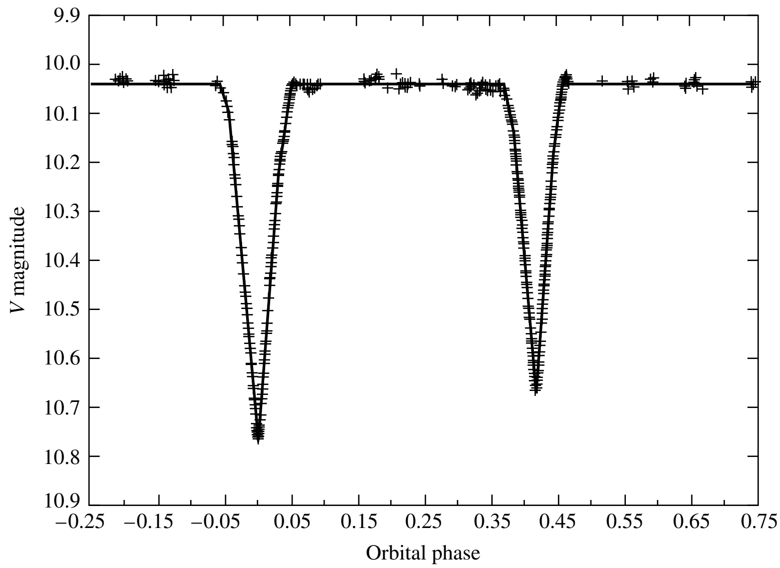

FIGURE 7.2 The V magnitude light curve of YY Sagittarii, an eclipsing binary star. The data from many orbital periods have been plotted on this light curve as a function of phase, where the phase is defined to be 0.0 at the primary minimum. This system has an orbital period $P = 2.6284734$ d, an eccentricity $e = 0.1573$, and orbital inclination $i = 88.89°$ (see Section 7.2). (Figure adopted from Lacy, C. H. S., *Astron. J.*, *105*, 637, 1993.)

Even if the Doppler shifts are not significant (if the orbital plane is perpendicular to the line of sight, for instance), it may still be possible to detect two sets of superimposed spectra if they originate from stars that have significantly different spectral features (see the discussion of spectral classes in Section 8.1).

- **Spectroscopic binary**. If the period of a binary system is not prohibitively long and if the orbital motion has a component along the line of sight, a periodic shift in spectral lines will be observable. Assuming that the luminosities of the stars are comparable, both spectra will be observable. However, if one star is much more luminous than the other, then the spectrum of the less luminous companion will be overwhelmed and only a single set of periodically varying spectral lines will be seen. In either situation, the existence of a binary star system is revealed. Figure 7.3 shows the relationship between spectra and orbital phase for a spectroscopic binary star system.

These specific classifications are not mutually exclusive. For instance, an unresolved system could be both an eclipsing and a spectroscopic binary. It is also true that some systems can be significantly more useful than others in providing information about stellar characteristics. Three types of systems can provide us with mass determinations: visual binaries combined with parallax information; visual binaries for which radial velocities are available over a complete orbit; and eclipsing, double-line, spectroscopic binaries.

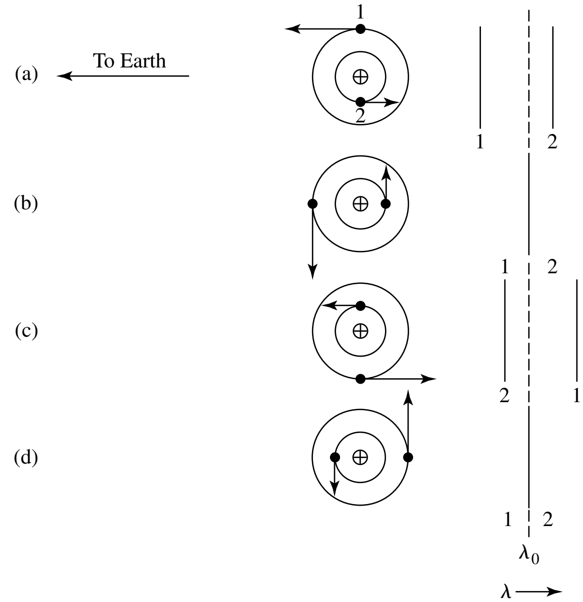

FIGURE 7.3 The periodic shift in spectral features of a double-line spectroscopic binary. The relative wavelengths of the spectra of Stars 1 and 2 are shown at four different phases during the orbit: (a) Star 1 is moving toward the observer while Star 2 is moving away. (b) Both stars have velocities perpendicular to the line of sight. (c) Star 1 is receding from the observer while Star 2 is approaching. (d) Again both stars have velocities perpendicular to the line of sight. λ_0 represents the wavelength of the observed line Doppler-shifted by the velocity of the center of mass of the system.

7.2 ■ MASS DETERMINATION USING VISUAL BINARIES

When the angular separation between components of a binary system is greater than the resolution limit imposed by local seeing conditions and the fundamental diffraction limitation of the Rayleigh criterion, it becomes possible to analyze the orbital characteristics of the individual stars. From the orbital data, the orientation of the orbits and the system's center of mass can be determined, providing knowledge of the ratio of the stars' masses. If the distance to the system is also known, from trigonometric parallax for instance, the linear separation of the stars can be determined, leading to the individual masses of the stars in the system.

To see how a visual binary can yield mass information, consider two stars in orbit about their mutual center of mass. Assuming that the orbital plane is perpendicular to the observer's line of sight, we see from the discussion of Section 2.3 that the ratio of masses may be found from the ratio of the angular separations of the stars from the center of mass. Using Eq. (2.19) and considering only the lengths of the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$, we find that

$$\frac{m_1}{m_2} = \frac{r_2}{r_1} = \frac{a_2}{a_1}, \tag{7.1}$$

where a_1 and a_2 are the semimajor axes of the ellipses. If the distance from the observer to the binary star system is d, then the angles subtended by the semimajor axes are

$$\alpha_1 = \frac{a_1}{d} \qquad \text{and} \qquad \alpha_2 = \frac{a_2}{d},$$

where α_1 and α_2 are measured in radians. Substituting, we find that the mass ratio simply becomes

$$\boxed{\frac{m_1}{m_2} = \frac{\alpha_2}{\alpha_1}.} \tag{7.2}$$

Even if the distance to the star system is not known, the mass ratio may still be determined. Note that since only the ratio of the subtended angles is needed, α_1 and α_2 may be expressed in arcseconds, the unit typically used for angular measure in astronomy.

The general form of Kepler's third law (Eq. 2.37),

$$P^2 = \frac{4\pi^2}{G\,(m_1 + m_2)} a^3,$$

gives the sum of the masses of the stars, provided that the semimajor axis (a) of the orbit of the reduced mass is known. Since $a = a_1 + a_2$ (the proof of this is left as an exercise), the semimajor axis can be determined directly only if the distance to the system has been determined. Assuming that d is known, $m_1 + m_2$ may be combined with m_1/m_2 to give each mass separately.

This process is complicated somewhat by the proper motion of the center of mass[1] (see Fig. 7.1) and by the fact that most orbits are not conveniently oriented with their planes perpendicular to the line of sight of the observer. Removing the proper motion of the center of mass from the observations is a relatively simple process since the center of mass must move at a constant velocity. Fortunately, estimating the orientation of the orbits is also possible and can be taken into consideration.

Let i be the **angle of inclination** between the plane of an orbit and the plane of the sky, as shown in Fig. 7.4; note that the orbits of both stars are necessarily in the same plane. As a special case, assume that the orbital plane and the plane of the sky (defined as being perpendicular to the line of sight) intersect along a line parallel to the minor axis, forming a **line of nodes**. The observer will not measure the actual angles subtended by the semimajor axes α_1 and α_2 but their projections onto the plane of the sky, $\tilde{\alpha}_1 = \alpha_1 \cos i$ and $\tilde{\alpha}_2 = \alpha_2 \cos i$. This geometrical effect plays no role in calculating the mass ratios since the $\cos i$ term will simply cancel in Eq. (7.2):

$$\frac{m_1}{m_2} = \frac{\alpha_2}{\alpha_1} = \frac{\alpha_2 \cos i}{\alpha_1 \cos i} = \frac{\tilde{\alpha}_2}{\tilde{\alpha}_1}.$$

However, this projection effect can make a significant difference when we are using Kepler's third law. Since $\alpha = a/d$ (α in radians), Kepler's third law may be solved for the sum of

[1]The annual wobble of stellar positions due to trigonometric parallax must also be considered, when significant.

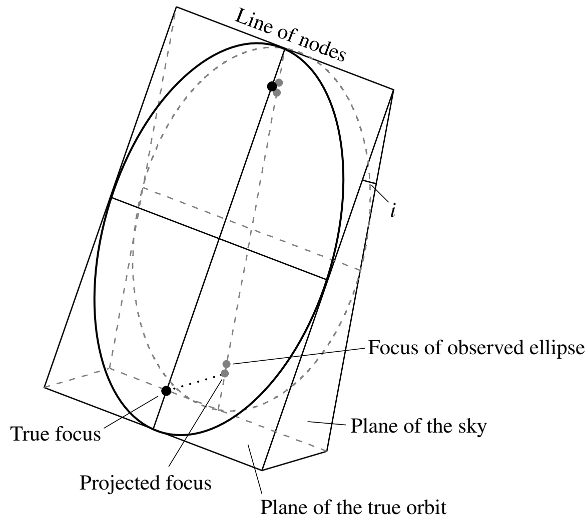

FIGURE 7.4 An elliptical orbit projected onto the plane of the sky produces an observable elliptical orbit. The foci of the original ellipse do not project onto the foci of the observed ellipse, however.

the masses to give

$$\boxed{m_1 + m_2 = \frac{4\pi^2}{G}\frac{(\alpha d)^3}{P^2} = \frac{4\pi^2}{G}\left(\frac{d}{\cos i}\right)^3 \frac{\tilde{\alpha}^3}{P^2},} \tag{7.3}$$

where $\tilde{\alpha} = \tilde{\alpha}_1 + \tilde{\alpha}_2$.

To evaluate the sum of the masses properly, we must deduce the angle of inclination. This can be accomplished by carefully noting the apparent position of the center of mass of the system. As illustrated in Fig. 7.4, the projection of an ellipse tilted at an angle i with respect to the plane of the sky will result in an observed ellipse with a different eccentricity. However, the center of mass will not be located at one of the foci of the projection—a result that is inconsistent with Kepler's first law. Thus the geometry of the true ellipse may be determined by comparing the observed stellar positions with mathematical projections of various ellipses onto the plane of the sky.

Of course, the problem of projection has been simplified here. Not only can the angle of inclination i be nonzero, but the ellipse may be tilted about its major axis and rotated about the line of sight to produce any possible orientation. However, the general principles already mentioned still apply, making it possible to deduce the true shapes of the stars' elliptical orbits, as well as their masses.

It is also possible to determine the individual masses of members of visual binaries, even if the distance is not known. In this situation, detailed radial velocity data are needed. The projection of velocity vectors onto the line of sight, combined with information about the stars' positions and the orientation of their orbits, provides a means for determining the semimajor axes of the ellipses, as required by Kepler's third law.

7.3 ■ ECLIPSING, SPECTROSCOPIC BINARIES

A wealth of information is available from a binary system even if it is not possible to resolve each of its stars individually. This is particularly true for a double-line, eclipsing, spectroscopic binary star system. In such a system, not only is it possible to determine the individual masses of the stars, but astronomers may be able to deduce other parameters as well, such as the stars' radii and the ratio of their fluxes, and hence the ratio of their effective temperatures. (Of course, eclipsing systems are not restricted to spectroscopic binaries but may occur in other types of binaries as well, such as visual binaries.)

The Effect of Eccentricity on Radial Velocity Measurements

Consider a spectroscopic binary star system for which the spectra of both stars may be seen (a double-line, spectroscopic binary). Since the individual members of the system cannot be resolved, the techniques used to determine the orientation and eccentricity of the orbits of visual binaries are not applicable. Also, the inclination angle i clearly plays a role in the solution obtained for the stars' masses because it directly influences the measured radial velocities. If v_1 is the velocity of the star of mass m_1 and v_2 is the velocity of the star of mass m_2 at some instant, then, referring to Fig. 7.4, the observed radial velocities cannot exceed $v_{1r}^{\max} = v_1 \sin i$ and $v_{2r}^{\max} = v_2 \sin i$, respectively. Therefore, the actual measured radial velocities depend upon the positions of the stars at that instant. As a special case, if the directions of motion of the stars happen to be perpendicular to the line of sight, then the observed radial velocities will be zero.

For a star system having circular orbits, the speed of each star will be constant. If the plane of their orbits lies in the line of sight of the observer ($i = 90°$), then the measured radial velocities will produce sinusoidal *velocity curves*, as in Fig. 7.5. Changing the orbital inclination does not alter the shape of the velocity curves; it merely changes their amplitudes

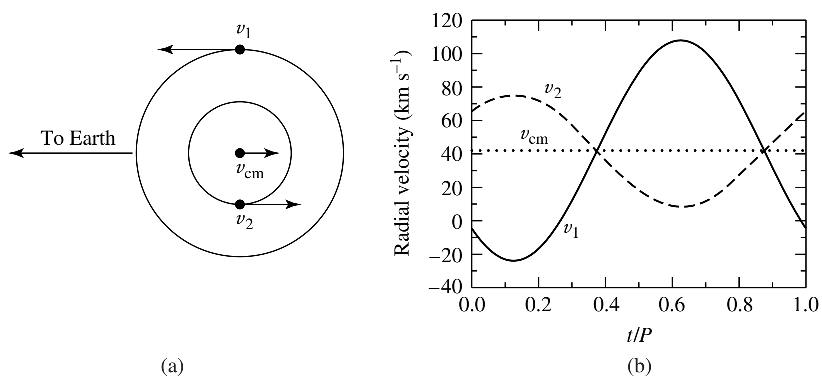

FIGURE 7.5 The orbital paths and radial velocities of two stars in circular orbits ($e = 0$). In this example, $M_1 = 1\ M_\odot$, $M_2 = 2\ M_\odot$, the orbital period is $P = 30$ d, and the radial velocity of the center of mass is $v_{cm} = 42$ km s^{-1}. v_1, v_2, and v_{cm} are the velocities of Star 1, Star 2, and the center of mass, respectively. (a) The plane of the circular orbits lies along the line of sight of the observer. (b) The observed radial velocity curves.

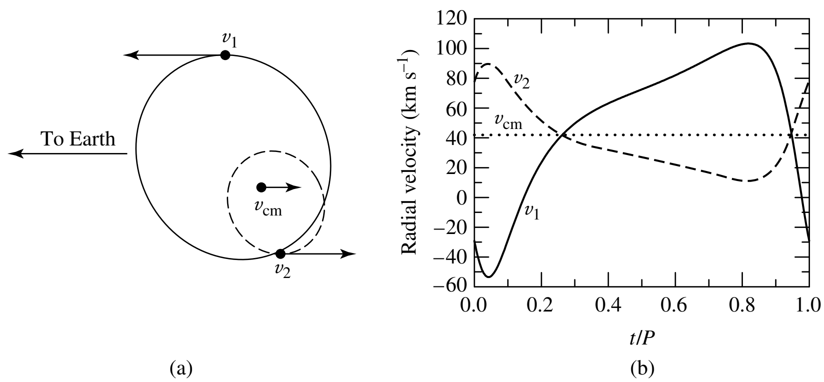

FIGURE 7.6 The orbital paths and radial velocities of two stars in elliptical orbits ($e = 0.4$). As in Fig. 7.5, $M_1 = 1\ \mathrm{M}_\odot$, $M_2 = 2\ \mathrm{M}_\odot$, the orbital period is $P = 30$ d, and the radial velocity of the center of mass is $v_{cm} = 42\ \mathrm{km\ s^{-1}}$. In addition, the orientation of periastron is 45°. v_1, v_2, and v_{cm} are the velocities of Star 1, Star 2, and the center of mass, respectively. (a) The plane of the orbits lies along the line of sight of the observer. (b) The observed radial velocity curves.

by the factor $\sin i$. To estimate i and the actual orbital velocities, therefore, other information about the system is necessary.

When the eccentricity, e, of the orbits is not zero, the observed velocity curves become skewed, as shown in Fig. 7.6. The exact shapes of the curves also depend strongly on the orientation of the orbits with respect to the observer, even for a given inclination angle.

In reality, many spectroscopic binaries possess nearly circular orbits, simplifying the analysis of the system somewhat. This occurs because close binaries tend to circularize their orbits due to tidal interactions over timescales that are short compared to the lifetimes of the stars involved.

The Mass Function and the Mass–Luminosity Relation

If we assume that the orbital eccentricity is very small ($e \ll 1$), then the speeds of the stars are essentially constant and given by $v_1 = 2\pi a_1/P$ and $v_2 = 2\pi a_2/P$ for stars of mass m_1 and m_2, respectively, where a_1 and a_2 are the radii (semimajor axes) and P is the period of the orbits. Solving for a_1 and a_2 and substituting into Eq. (7.1), we find that the ratio of the masses of the two stars becomes

$$\frac{m_1}{m_2} = \frac{v_2}{v_1}. \tag{7.4}$$

Because $v_{1r} = v_1 \sin i$ and $v_{2r} = v_2 \sin i$, Eq. (7.4) can be written in terms of the observed radial velocities rather than actual orbital velocities:

$$\boxed{\frac{m_1}{m_2} = \frac{v_{2r}/\sin i}{v_{1r}/\sin i} = \frac{v_{2r}}{v_{1r}}.} \tag{7.5}$$

As is the situation with visual binaries, we can determine the ratio of the stellar masses without knowing the angle of inclination.

However, as is also the case with visual binaries, finding the sum of the masses does require knowledge of the angle of inclination. Replacing a with

$$a = a_1 + a_2 = \frac{P}{2\pi}\,(v_1 + v_2)$$

in Kepler's third law (Eq. 2.37) and solving for the sum of the masses, we have

$$m_1 + m_2 = \frac{P}{2\pi G}\,(v_1 + v_2)^3 .$$

Writing the actual radial velocities in terms of the observed values, we can express the sum of the masses as

$$\boxed{m_1 + m_2 = \frac{P}{2\pi G}\frac{(v_{1r} + v_{2r})^3}{\sin^3 i}.} \tag{7.6}$$

It is clear from Eq. (7.6) that the sum of the masses can be obtained only if both v_{1r} and v_{2r} are measurable. Unfortunately, this is not always the case. If one star is much brighter than its companion, the spectrum of the dimmer member will be overwhelmed. Such a system is referred to as a *single-line spectroscopic binary*. If the spectrum of Star 1 is observable but the spectrum of Star 2 is not, Eq. (7.5) allows v_{2r} to be replaced by the ratio of the stellar masses, giving a quantity that is dependent on both of the system masses and the angle of inclination. If we substitute, Eq. (7.6) becomes

$$m_1 + m_2 = \frac{P}{2\pi G}\frac{v_{1r}^3}{\sin^3 i}\left(1 + \frac{m_1}{m_2}\right)^3 .$$

Rearranging terms gives

$$\boxed{\frac{m_2^3}{(m_1 + m_2)^2}\sin^3 i = \frac{P}{2\pi G}\,v_{1r}^3.} \tag{7.7}$$

The right-hand side of this expression, known as the **mass function**, depends only on the readily observable quantities, period and radial velocity. Since the spectrum of only one star is available, Eq. (7.5) cannot provide any information about mass ratios. As a result, the mass function is useful only for statistical studies or if an estimate of the mass of at least one component of the system already exists by some indirect means. If either m_1 or $\sin i$ is unknown, the mass function sets a lower limit for m_2, since the left-hand side is always less than m_2.

Even if both radial velocities are measurable, it is not possible to get exact values for m_1 and m_2 without knowing i. However, since stars can be grouped according to their effective temperatures and luminosities (see Section 8.2), and assuming that there is a relationship between these quantities and mass, then a statistical mass estimate for each class may be found by choosing an appropriately averaged value for $\sin^3 i$. An integral average of $\sin^3 i$ ($\langle \sin^3 i\rangle$) evaluated between 0° and 90° has a value $3\pi/16 \simeq 0.589$.[2] However, since no

[2]The proof is left as an exercise.

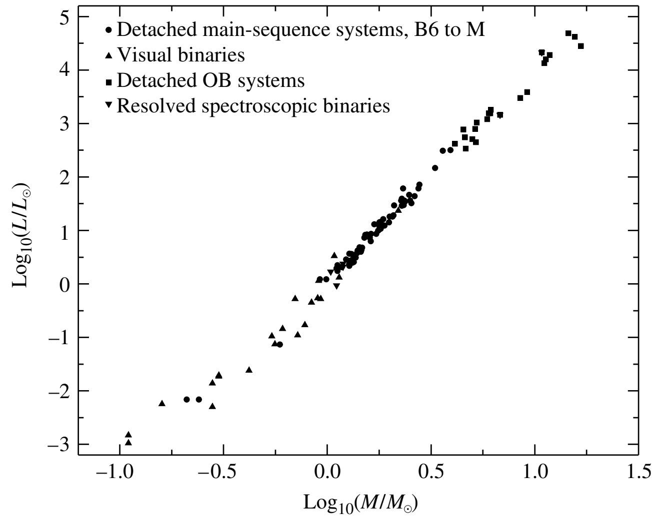

FIGURE 7.7 The mass–luminosity relation. (Data from Popper, *Annu. Rev. Astron. Astrophys.*, *18*, 115, 1980.)

Doppler shift will be noticeable if the inclination angle is very small, it is more likely that a spectroscopic binary star system will be discovered if i differs significantly from 0°. This *selection effect* associated with detecting binary systems suggests that a larger value of $\langle \sin^3 i \rangle \simeq 2/3$ is more representative.

Evaluating masses of binaries has shown the existence of a well-defined **mass–luminosity relation** for the large majority of stars in the sky (see Fig. 7.7). One of the goals of the next several chapters is to investigate the origin of this relation in terms of fundamental physical principles.

Using Eclipses to Determine Radii and Ratios of Temperatures

A good estimate of i is possible in the special situation that a spectroscopic binary star system is observed to be an eclipsing system as well. Unless the distance of separation between the components of the binary is not much larger than the sum of the radii of the stars involved, an eclipsing system implies that i must be close to 90°, as suggested in Fig. 7.8. Even if it were assumed that $i = 90°$ while the actual value was closer to 75°, an error of only 10% would result in the calculation of $\sin^3 i$ and in the determination of $m_1 + m_2$.

From the light curves produced by eclipsing binaries, it is possible to improve the estimate of i still further. Figure 7.9 indicates that if the smaller star is completely eclipsed by the larger one, a nearly constant minimum will occur in the measured brightness of the system during the period of occultation. Similarly, even though the larger star will not be fully hidden from view when the smaller companion passes in front of it, a constant amount of area will still be obscured for a time, and again a nearly constant, though diminished amount of light will be observed. When one star is not completely eclipsed by its companion (Fig. 7.10), the minima are no longer constant, implying that i must be less than 90°.

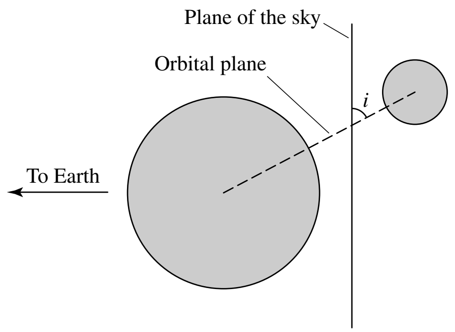

FIGURE 7.8 The geometry of an eclipsing, spectroscopic binary requires that the angle of inclination i be close to 90°.

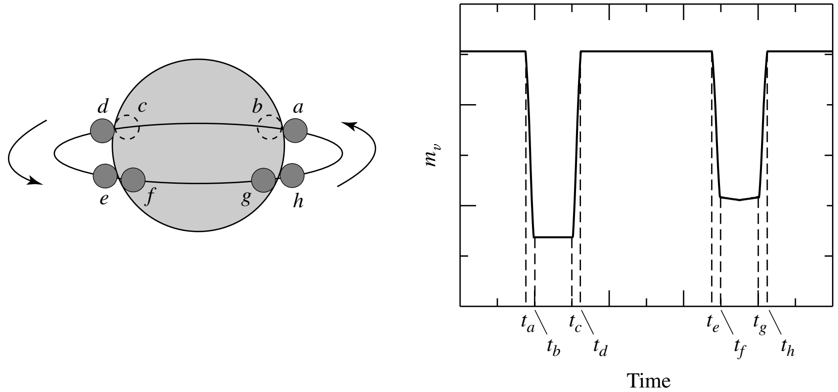

FIGURE 7.9 The light curve of an eclipsing binary for which $i = 90°$. The times indicated on the light curve correspond to the positions of the smaller star relative to its larger companion. It is assumed in this example that the smaller star is hotter than the larger one.

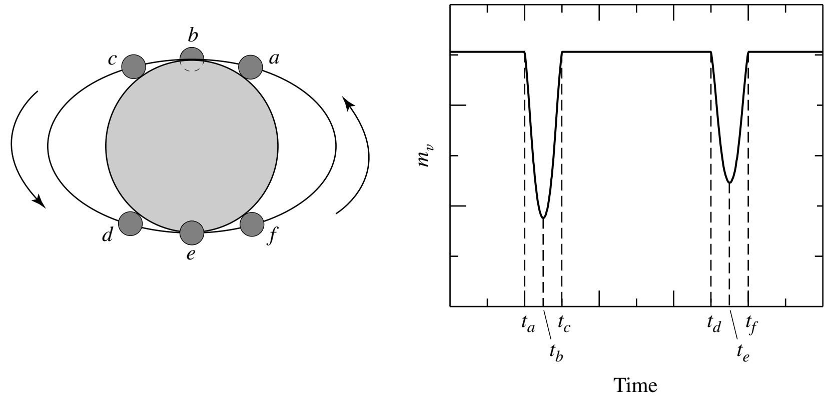

FIGURE 7.10 The light curve of a partially eclipsing binary. It is assumed in this example that the smaller star is hotter than its companion.

Using measurements of the duration of eclipses, it is also possible to find the radii of each member of an eclipsing, spectroscopic binary. Referring again to Fig. 7.9, if we assume that $i \simeq 90°$, the amount of time between *first contact* (t_a) and minimum light (t_b), combined with the velocities of the stars, leads directly to the calculation of the radius of the smaller component. For example, if the semimajor axis of the smaller star's orbit is sufficiently large compared to either star's radius, and if the orbit is nearly circular, we can assume that the smaller object is moving approximately perpendicular to the line of sight of the observer during the duration of the eclipse. In this case the radius of the smaller star is simply

$$r_s = \frac{v}{2}\left(t_b - t_a\right), \tag{7.8}$$

where $v = v_s + v_\ell$ is the *relative* velocity of the two stars (v_s and v_ℓ are the velocities of the small and large stars, respectively). Similarly, if we consider the amount of time between t_b and t_c, the size of the larger member can also be determined. It can be quickly shown that the radius of the larger star is just

$$r_\ell = \frac{v}{2}\left(t_c - t_a\right) = r_s + \frac{v}{2}\left(t_c - t_b\right). \tag{7.9}$$

Example 7.3.1. An analysis of the spectrum of an eclipsing, double-line, spectroscopic binary having a period of $P = 8.6$ yr shows that the maximum Doppler shift of the hydrogen Balmer Hα (656.281 nm) line is $\Delta\lambda_s = 0.072$ nm for the smaller member and only $\Delta\lambda_\ell = 0.0068$ nm for its companion. From the sinusoidal shapes of the velocity curves, it is also apparent that the orbits are nearly circular. Using Eqs. (4.39) and (7.5), we find that the mass ratio of the two stars must be

$$\frac{m_\ell}{m_s} = \frac{v_{rs}}{v_{r\ell}} = \frac{\Delta\lambda_s}{\Delta\lambda_\ell} = 10.6.$$

Assuming that the orbital inclination is $i = 90°$, the Doppler shift of the smaller star implies that the maximum measured radial velocity is

$$v_{rs} = \frac{\Delta\lambda_s}{\lambda}c = 33 \text{ km s}^{-1}$$

and the radius of its orbit must be

$$a_s = \frac{v_{rs}P}{2\pi} = 1.42 \times 10^{12} \text{ m} = 9.5 \text{ AU}.$$

In the same manner, the orbital velocity and radius of the other star are $v_{r\ell} = 3.1$ km s^{-1} and $a_\ell = 0.90$ AU, respectively. Therefore, the semimajor axis of the reduced mass becomes $a = a_s + a_\ell = 10.4$ AU.

continued

The sum of the masses can now be determined from Kepler's third law. If Eq. (2.37) is written in units of solar masses, astronomical units, and years, we have

$$m_s + m_\ell = a^3/P^2 = 15.2\ \mathrm{M}_\odot.$$

Solving for the masses independently yields $m_s = 1.3\ \mathrm{M}_\odot$ and $m_\ell = 13.9\ \mathrm{M}_\odot$.

Furthermore, from the light curve for this system, it is found that $t_b - t_a = 11.7$ hours and $t_c - t_b = 164$ days. Using Eq. (7.8) reveals that the radius of the smaller star is

$$r_s = \frac{(v_{rs} + v_{r\ell})}{2}(t_b - t_a) = 7.6 \times 10^8\ \mathrm{m} = 1.1\ \mathrm{R}_\odot,$$

where one solar radius is $1\ \mathrm{R}_\odot = 6.96 \times 10^8$ m. Equation (7.9) now gives the radius of the larger star, which is found to be $r_\ell = 369\ \mathrm{R}_\odot$.

In this particular system, the masses and radii of the stars are found to differ significantly.

The ratio of the effective temperatures of the two stars can also be obtained from the light curve of an eclipsing binary. This is accomplished by considering the objects as blackbody radiators and comparing the amount of light received during an eclipse with the amount received when both members are fully visible.

Referring once more to the sample binary system depicted in Fig. 7.9, it can be seen that the dip in the light curve is deeper when the smaller, hotter star is passing behind its companion. To understand this effect, recall that the radiative surface flux is given by Eq. (3.18),

$$F_r = F_{\mathrm{surf}} = \sigma T_e^4.$$

Regardless of whether the smaller star passes behind or in front of the larger one, the same total cross-sectional area is eclipsed. Assuming for simplicity that the observed flux is constant across the disks,[3] the amount of light detected from the binary when both stars are fully visible is given by

$$B_0 = k\left(\pi r_\ell^2 F_{r\ell} + \pi r_s^2 F_{rs}\right),$$

where k is a constant that depends on the distance to the system, the amount of intervening material between the system and the detector, and the nature of the detector. The deeper, or *primary*, minimum occurs when the hotter star passes behind the cooler one. If, as in the last example, the smaller star is hotter and therefore has the larger surface flux, and the smaller star is entirely eclipsed, the amount of light detected during the primary minimum may be expressed as

$$B_p = k\pi r_\ell^2 F_{r\ell}$$

while the brightness of the *secondary* minimum is

$$B_s = k\left(\pi r_\ell^2 - \pi r_s^2\right) F_{r\ell} + k\pi r_s^2 F_{rs}.$$

[3] Stars often appear darker near the edges of their disks, a phenomenon referred to as *limb darkening*. This effect will be discussed in Section 9.3.

Since it is generally not possible to determine k exactly, ratios are employed. Consider the ratio of the depth of the primary to the depth of the secondary. Using the expressions for B_0, B_p, and B_s, we find immediately that

$$\frac{B_0 - B_p}{B_0 - B_s} = \frac{F_{rs}}{F_{r\ell}} \tag{7.10}$$

or, from Eq. (3.18),

$$\boxed{\frac{B_0 - B_p}{B_0 - B_s} = \left(\frac{T_s}{T_\ell}\right)^4.} \tag{7.11}$$

Example 7.3.2. Further examination of the light curve of the binary system discussed in Example 7.3.1 provides information on the relative temperatures of the two stars. Photometric observations show that at maximum light the bolometric magnitude is $m_{\text{bol},0} = 6.3$, at the primary minimum $m_{\text{bol},p} = 9.6$, and at the secondary minimum $m_{\text{bol},s} = 6.6$. From Eq. (3.3), the ratio of brightnesses between the primary minimum and maximum light is

$$\frac{B_p}{B_0} = 100^{(m_{\text{bol},0} - m_{\text{bol},p})/5} = 0.048.$$

Similarly, the ratio of brightnesses between the secondary minimum and maximum light is

$$\frac{B_s}{B_0} = 100^{(m_{\text{bol},0} - m_{\text{bol},s})/5} = 0.76.$$

Now, by rewriting Eq. (7.10), we find that the ratio of the radiative fluxes is

$$\frac{F_{rs}}{F_{r\ell}} = \frac{1 - B_p/B_0}{1 - B_s/B_0} = 3.97.$$

Finally, from Eq. (3.18),

$$\frac{T_s}{T_\ell} = \left(\frac{F_{rs}}{F_{r\ell}}\right)^{1/4} = 1.41.$$

A Computer Modeling Approach

The modern approach to analyzing the data from binary star systems involves computing detailed models that can yield important information about a variety of physical parameters. Not only can masses, radii, and effective temperatures be determined, but for many systems other details can be described as well. For instance, gravitational forces, combined with the effects of rotation and orbital motion, alter the stars' shapes; they are no longer simply spherical objects but may become elongated (these effects will be discussed in more detail in Section 18.1). The models may also incorporate information about the nonuniform

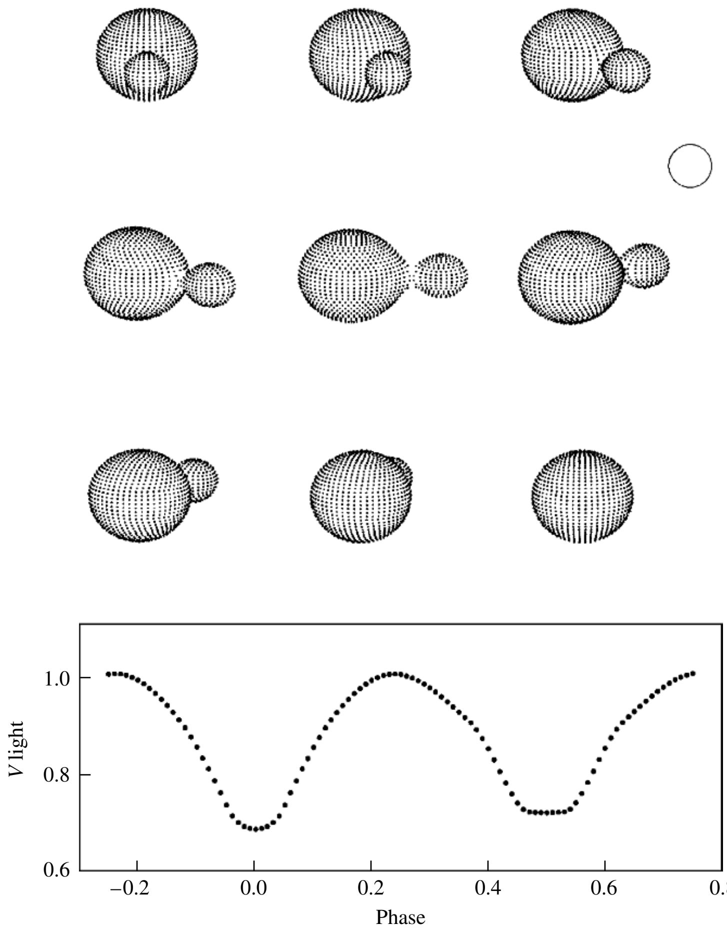

FIGURE 7.11 A synthetic light curve of RR Centauri, an eclipsing binary star system for which the two components are in close contact. The open circle represents the size of the Sun. The orbital and physical characteristics of the RR Cen system are $P = 0.6057$ d, $e = 0.0$, $M_1 = 1.8\ \mathrm{M}_\odot$, $M_2 = 0.37\ \mathrm{M}_\odot$. The spectral classification of the primary is F0V (see Section 8.1 for a discussion of stellar spectral classifications). (Figure adapted from R. E. Wilson, *Publ. Astron. Soc. Pac.*, *106*, 921, 1994; ©Astronomical Society of the Pacific.)

distribution of flux across the observed disks of the stars, variations in surface temperatures, and so on. Once the shapes of the gravitational equipotential surfaces and other parameters are determined, *synthetic* (theoretical) light curves can be computed for various wavelength bands (U, B, V, etc.), which are then compared to the observational data. Adjustments in the model parameters are made until the light curves agree with the observations. One such model for the binary system RR Centauri is shown in Fig. 7.11. In this system the two stars are actually in contact with each other, producing interesting and subtle effects in the light curve.

In order to introduce you to the process of modeling binary systems, the simple code `TwoStars` is described in Appendix K and available on the companion website. `TwoStars` makes the simplifying assumption that the stars are perfectly spherically symmetric. Thus `TwoStars` is capable of generating light curves, radial velocity curves, and astrometric data for systems in which the two stars are well separated. The simplifying assumptions imply that `TwoStars` is incapable of modeling the details of more complicated systems such as RR Cen, however.[4]

The study of binary star systems provides valuable information about the observable characteristics of stars. These results are then employed in developing a theory of stellar structure and evolution.

7.4 ■ THE SEARCH FOR EXTRASOLAR PLANETS

For hundreds of years, people have looked up at the night sky and wondered if planets might exist around other stars.[5] However, it wasn't until October 1995 that Michel Mayor and Didier Queloz of the Geneva Observatory announced the discovery of a planet around the solar-type star 51 Pegasi. This discovery represented the first detection of an extrasolar planet around a typical star.[6] Within one month of the announced discovery of 51 Peg, Geoffery W. Marcy and R. Paul Butler of the University of California, Berkeley, and the Carnegie Institution of Washington, respectively, announced that they had detected planets around two other Sun-like stars, 70 Vir and 47 UMa. By May 2006, just over ten years after the original announcements, 189 extrasolar planets had been discovered orbiting 163 stars that are similar to our own Sun.

This modern discovery of extrasolar planets at such a prodigious rate was made possible by dramatic advances in detector technology, the availability of large-aperture telescopes, and diligent, long-term observing campaigns. Given the huge disparity between the luminosity of the parent star and any orbiting planets, direct observation of a planet has proved very difficult; the planet's reflected light is simply overwhelmed by the luminosity of the star.[7] As a result, more indirect methods are usually required to detect extrasolar planets. Three techniques that have all been used successfully are based on ideas discussed in this chapter: radial velocity measurements, astrometric wobbles, and eclipses.[8] The first method, the detection of radial velocity variations in parent stars induced by the gravitational tug of the orbiting planets has been by far the most prolific method at the time this text was written.

[4]More sophisticated binary star modeling codes are available for download on the Internet or may be purchased. Examples include `WD95`, originally written by Wilson and Devinney and later modified by Kallrath, et al., and `Binary Maker` by Bradstreet and Steelman.

[5]In fact, it is thought that Giordano Bruno (1548–1600), a one-time Dominican monk, was executed for his belief in a Copernican universe filled with an infinite number of inhabited worlds around other stars; recall Section 1.2.

[6]In 1992, Alexander Wolszczan, of the Arecibo Radio Observatory in Puerto Rico, and Dale Frail, of the National Radio Astronomy Observatory, detected three Earth- and Moon-sized planets around a pulsar (PSR 1257+12), an extremely compact collapsed star that was produced following a supernova explosion; see Section 16.7. This discovery was made by noting variations in the extremely regular radio emission coming from the collapsed star.

[7]In April 2004, G. Chauvin and colleagues used the VLT/NACO of the European Southern Observatory to obtain an infrared image of a giant extrasolar planet of spectral type between L5 and L9.5 orbiting the brown dwarf 2MASSWJ1207334–393254. HST/NICMOS was also able to observe the brown dwarf's planetary companion.

[8]Another technique has also been employed in the search for extrasolar planets; it is based on the gravitational lensing of light; see Chapter 17.

Example 7.4.1. The so-called *reflex motion* of the parent star is extremely small. For example, consider the motion of Jupiter around the Sun. Jupiter's orbital period is 11.86 yr, the semimajor axis of its orbit is 5.2 AU, and its mass is only 0.000955 $M_\odot$. Assuming that the orbit of Jupiter is essentially circular (its actual eccentricity is just $e = 0.0489$), the planet's orbital velocity is approximately

$$v_J = 2\pi a/P = 13.1 \text{ km s}^{-1}.$$

According to Eq. (7.5), the Sun's orbital velocity about their mutual center of mass is only

$$v_\odot = \frac{m_J}{M_\odot} v_J = 12.5 \text{ m s}^{-1}.$$

This is similar to the top speed of a world-class sprinter from Earth.

Incredibly, today it is possible to measure radial velocity variations as small as 3 m s^{-1}, a slow jog in the park. Marcy, Butler, and their research-team colleagues accomplish this level of detection by passing starlight through an iodine vapor. The imprinted absorption lines from the iodine are used as zero-velocity reference lines in the high-resolution spectrum of the star. By comparing the absorption and emission-line wavelengths of the star to the iodine reference wavelengths, it is possible to determine very precise radial velocities. The high-resolution spectrographs used by the team were designed and built by another team member, Steve Vogt of the University of California, Santa Cruz.

The analysis of the radial velocities requires much more work before the true reflex motion radial velocity variations of the star can be deduced, however. In order to determine the source of the variations, it is first necessary to eliminate all other sources of radial velocities superimposed on the observed spectra. These include the rotation and wobble of Earth, the orbital velocity of Earth around the Sun, and the gravitational effects of the other planets in our Solar System on Earth and our Sun. After all of these corrections have been made, the radial velocity of the target star can be referenced to the true center of mass of our Solar System.

In addition to the motions in our Solar System, motions of the target star itself must be taken into account. For instance, if the target star is rotating, radial velocities due to the approaching and receding edges of its apparent disk will blur the absorption lines used to measure radial velocity. Pulsations of the surface of the star (see Chapter 14), surface convection (Section 10.4), and the movement of surface features such as star spots (e.g., Section 11.3), can also confuse the measurements and degrade the velocity resolution limit.

All of the planets discovered by the radial velocity technique are quite close to their parent star and very massive. For instance, the lower limit for the mass of the planet orbiting 51 Peg is 0.45 M_J (where M_J is the mass of Jupiter), it has an orbital period of just 4.23077 d, and the semimajor axis of its orbit is only 0.051 AU. The lower limit on the mass of the planet orbiting HD 168443c is 16.96 M_J, its orbital period is 1770 d, and the semimajor axis of its orbit is 2.87 AU. As the length of time that stars are observed increases, longer orbital-period planets will continue to be discovered, as will lower-mass planets.

Careful analysis of the radial velocity curves of one star, HD 209458, led researchers to predict and then detect transits of an extrasolar planet across the star's disk in 1999. The dimming of the light due to the transits is completely analogous to the eclipsing,

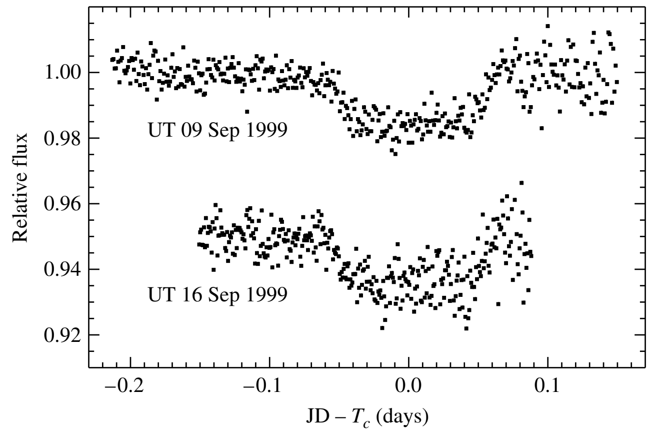

FIGURE 7.12 The photometric detection of two transits of an extrasolar planet across the disk of HD 2094589 in September 1999. The September 16 transit was artificially offset by −0.05 relative to the transit of September 9 in order to avoid overlap of the data. T_c designates the midpoint of the transit, and JD represents the Julian Date (time) of the particular measurement. (Figure adapted from Charbonneau, Brown, Latham, and Mayor, *Ap. J., 529*, L45, 2000.)

spectroscopic binary star systems discussed in Section 7.3. Given the very small size of the planet relative to HD 209458, the dimming of the light was only about 4 mmag (millimagnitudes); see Fig. 7.12. Based on the additional information provided by the light curves during the transits, Charbonneau, Brown, Latham, and Mayor were able to determine that the transiting planet has a radius of approximately 1.27 R_J (Jupiter radii) and that the orbital inclination $i = 87.1° \pm 0.2°$. Having restricted the value of the inclination angle, it then became possible to largely remove the uncertainty of the $\sin i$ term, resulting in a mass determination for the planet of 0.63 M_J. From the radial velocity data, the mass and radius of HD 209458 are 1.1 $M_\odot$ and 1.1 $R_\odot$, respectively. The orbital period of the planet is $P = 3.5250 \pm 0.003$ d, and the semimajor axis of its orbit is $a = 0.0467$ AU.

To date a number of planets have been detected by the dimming of starlight resulting from their transits of the disks of their parent stars. However, OGLE-TR-56b was the first system for which a planet was detected before it was found by the radial velocity technique. The orbital period of the planet is only 29 h and it orbits just 4.5 stellar radii (0.023 AU) from its parent star. The measurement was made by detecting a drop of slightly more than 0.01 magnitudes in the brightness of the star. An advantage of this technique is the ability to detect relatively distant systems; OGLE-TR-56b is approximately 1500 pc from Earth. In addition, from the transit time, the radius of the planet can also be determined, enabling an estimate of the density of the planet. The mass of the planet orbiting OGLE-TR-56b is estimated to be 0.9 M_J, as confirmed by follow-up radial velocity measurements, and its radius is just slightly larger than Jupiter's.

The reflex motion of a star due to the pull of a planet was detected for the first time in 2002. The Hubble Space Telescope's Fine Guidance Sensors were used to measure 0.5-milliarcsecond wobbles in Gliese 876, a tenth-magnitude star located 4.7 pc from Earth. By adding the third dimension of projection onto the plane of the sky, the mass of the planet previously obtained by the radial velocity technique was refined to give a value between

1.9 M_J and 2.4 M_J. When future astrometric missions are launched (see Section 6.5), it is likely that this technique will result in detection of many more planets.

Although Earth-sized planets have yet to be discovered around solar-type stars, with missions such as NASA's Terrestrial Planet Finder being planned, and exquisitely sensitive astrometric missions such as SIM PlanetQuest and Gaia, it seems likely that such discoveries will occur soon.

SUGGESTED READING

General

Burnham, Robert Jr., *Burnham's Celestial Handbook: An Observer's Guide to the Universe Beyond the Solar System*, Revised and Enlarged Edition, Dover Publications, New York, 1978.

Jones, Kenneth Glyn (ed.), *Webb Society Deep-Sky Observer's Handbook*, Second Edition, Enslow Publishers, Hillside, NJ, 1986.

Marcy, Geoffrey, and Butler, R. Paul, "New Worlds: The Diversity of Planetary Systems," *Sky and Telescope*, March 1998.

Marcy, Geoffrey, et al., *California and Carnegie Planet Search Web Site*, `http://exoplanets.org`.

NASA, *Planet Quest: The Search for Another Earth Web Site*, `http://planetquest.jpl.nasa.gov`.

Pasachoff, Jay M., *Field Guide to the Stars and Planets*, Fourth Edition, Houghton Mifflin, Boston, 2000.

Technical

Batten, Alan H., Fletcher, J. Murray, and MacCarthy, D. G., "Eighth Catalogue of the Orbital Elements of Spectroscopic Binary Systems," *Publications of the Dominion Astrophysical Observatory*, *17*, 1989.

Böhm-Vitense, Erika, *Introduction to Stellar Astrophysics: Basic Stellar Observations and Data*, Volume 1, Cambridge University Press, Cambridge, 1989.

Bradstreet, D. H., and Steelman, D. P., "Binary Maker 3.0—An Interactive Graphics-Based Light Curve Synthesis Program Written in Java," *Bulletin of the American Astronomical Society*, January 2003.

Charbonneau, David, Brown, Timothy M., Latham, David W., and Mayor, Michel, "Detection of Planetary Transits Across a Sun-Like Star," *The Astrophysical Journal*, *529*, L45, 2000.

Eggen, O. J., "Masses of Visual Binary Stars," *Annual Review of Astronomy and Astrophysics*, *5*, 105, 1967.

Kallrath, Josef, and Milone, Eugene F., *Eclipsing Binary Stars: Modeling and Analysis*, Springer–Verlag, New York, 1999.

Kallrath, J, Milone, E. F., Terrell, D., and Young, A. T., "Recent Improvements to a Version of the Wilson–Devinney Program," *The Astrophysical Journal Supplement Series*, *508*, 308, 1998.

Kitchin, C. R., *Astrophysical Techniques*, Third Edition, Institute of Physics Publications, Philadelphia, 1998.

Marcy, Geoffrey W., and Butler, R. Paul, " Detection of Extrasolar Giant Planets," *Annual Review of Astronomy and Astrophysics*, *36*, 57, 1998.

Mayor, M., and Queloz, D., "A Jupiter-Mass Companion to a Solar-Type Star," *Nature*, *378*, 355, 1995.

Popper, Daniel M., "Determination of Masses of Eclipsing Binary Stars," *Annual Review of Astronomy and Astrophysics*, *5*, 85, 1967.

Popper, Daniel M., "Stellar Masses," *Annual Review of Astronomy and Astrophysics*, *18*, 115, 1980.

Wilson, R. E., "Binary-Star Light-Curve Models," *Publications of the Astronomical Society of the Pacific*, *106*, 921, 1994.

PROBLEMS

7.1 Consider two stars in orbit about a mutual center of mass. If a_1 is the semimajor axis of the orbit of star of mass m_1 and a_2 is the semimajor axis of the orbit of star of mass m_2, prove that the semimajor axis of the orbit of the *reduced mass* is given by $a = a_1 + a_2$. *Hint:* Review the discussion of Section 2.3 and recall that $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$.

7.2 In Problem 2.9 we discussed integral averages that implicitly assumed a probability distribution (or *weighting function*) that was constant throughout the interval over which the integral was applied. When a *normalized* weighting function $w(\tau)$ is considered, such that

$$\int_0^\tau w(\tau)\,d\tau = 1,$$

then the integral average of $f(\tau)$ becomes

$$\langle f(\tau)\rangle = \int_0^\tau f(\tau)\,w(\tau)\,d\tau.$$

Comparison with Problem 2.9 reveals that the weighting function implicitly used in that case was $w(\tau) = 1/\tau$ over the interval 0 to τ.

In evaluating $\langle \sin^3 i\rangle$ between 0 rad and $\pi/2$ rad (0° and 90°, respectively) as discussed on page 188, it is more likely that the radial velocity variations will be detected if the plane of the orbit is oriented along the line of sight. The weighting function should therefore take into consideration the projection of the plane of the orbital velocity onto the line of sight.

(a) Select an appropriate weighting function and show that your weighting function is normalized over the interval $i = 0$ to $\pi/2$.

(b) Prove that $\langle \sin^3 i\rangle = 3\pi/16$.

7.3 Assume that two stars are in circular orbits about a mutual center of mass and are separated by a distance a. Assume also that the angle of inclination is i and their stellar radii are r_1 and r_2.

(a) Find an expression for the smallest angle of inclination that will just barely produce an eclipse. *Hint:* Refer to Fig. 7.8.

(b) If $a = 2$ AU, $r_1 = 10\,\mathrm{R}_\odot$, and $r_2 = 1\,\mathrm{R}_\odot$, what minimum value of i will result in an eclipse?

7.4 Sirius is a visual binary with a period of 49.94 yr. Its measured trigonometric parallax is $0.37921'' \pm 0.00158''$ and, assuming that the plane of the orbit is in the plane of the sky, the true angular extent of the semimajor axis of the reduced mass is $7.61''$. The ratio of the distances of Sirius A and Sirius B from the center of mass is $a_A/a_B = 0.466$.

(a) Find the mass of each member of the system.

(b) The absolute bolometric magnitude of Sirius A is 1.36, and Sirius B has an absolute bolometric magnitude of 8.79. Determine their luminosities. Express your answers in terms of the luminosity of the Sun.

(c) The effective temperature of Sirius B is approximately 24,790 K $\pm$100 K. Estimate its radius, and compare your answer to the radii of the Sun and Earth.

7.5 ζ Phe is a 1.67-day spectroscopic binary with nearly circular orbits. The maximum measured Doppler shifts of the brighter and fainter components of the system are 121.4 km s^{-1} and 247 km s^{-1}, respectively.

(a) Determine the quantity $m \sin^3 i$ for each star.

(b) Using a statistically chosen value for $\sin^3 i$ that takes into consideration the Doppler-shift selection effect, estimate the individual masses of the components of ζ Phe.

7.6 From the light and velocity curves of an eclipsing, spectroscopic binary star system, it is determined that the orbital period is 6.31 yr, and the maximum radial velocities of Stars A and B are 5.4 km s^{-1} and 22.4 km s^{-1}, respectively. Furthermore, the time period between first contact and minimum light $(t_b - t_a)$ is 0.58 d, the length of the primary minimum $(t_c - t_b)$ is 0.64 d, and the apparent bolometric magnitudes of maximum, primary minimum, and secondary minimum are 5.40 magnitudes, 9.20 magnitudes, and 5.44 magnitudes, respectively. From this information, and assuming circular orbits, find the

(a) Ratio of stellar masses.

(b) Sum of the masses (assume $i \simeq 90°$).

(c) Individual masses.

(d) Individual radii (assume that the orbits are circular).

(e) Ratio of the effective temperatures of the two stars.

7.7 The V-band light curve of YY Sgr is shown in Fig. 7.2. Neglecting bolometric corrections, estimate the ratio of the temperatures of the two stars in the system.

7.8 Refer to the synthetic light curve and model of RR Centauri shown in Fig. 7.11.

(a) Indicate the approximate points on the light curve (as a function of phase) that correspond to the orientations depicted.

(b) Explain qualitatively the shape of the light curve.

7.9 Data from binary star systems were used to illustrate the mass–luminosity relation in Fig. 7.7. A strong correlation also exists between mass and the effective temperatures of stars. Use the data provided in Popper, *Annu. Rev. Astron. Astrophys.*, *18*, 115, 1980 to create a graph of $\log_{10} T_e$ as a function of $\log_{10}(M/M_\odot)$. Use the data from Popper's Table 2, Table 4, Table 7 (excluding the α Aur system), and Table 8 (include only those stars with spectral types in the Sp column that end with the Roman numeral V). The stars that are excluded in Tables 7 and 8 are evolved stars with structures significantly different from the main sequence stars.[9] The article by Popper may be available in your library or it can be downloaded from the NASA Astrophysics Data System (NASA ADS) at `http://adswww.harvard.edu`.

[9]Spectral types and different classes of stars will be discussed in detail in Chapter 8.

7.10 Give two reasons why the radial velocity technique for detecting planets around other stars favors massive planets (*Jupiters*) with relatively short orbital periods.

7.11 Explain why radial velocity detections of extrasolar planets yield only lower limits on the masses of the orbiting planets. What value is actually measured, and what unknown orbital parameter is involved?

7.12 From the data given in the text, determine the masses of the following stars (in solar masses):

(a) 51 Peg

(b) HD 168443c

7.13 Suppose that you are an astronomer on a planet orbiting another star. While you are observing our Sun, Jupiter passes in front of it. Estimate the fractional decrease in the brightness of the star, assuming that you are observing a flat disk of constant flux, with a temperature of $T_e = 5777$ K. *Hint:* Neglect Jupiter's contribution to the total brightness of the system.

7.14 From the data given in the text, combined with the information in Fig. 7.12, make a rough estimate of the radius of the orbiting planet, and compare your result with the quoted value. Be sure to explain each step used in computing your estimate.

COMPUTER PROBLEMS

7.15 **(a)** Use the computer program `TwoStars`, described in Appendix K and available on the companion website, to generate orbital radial velocity data similar to Fig. 7.6 for any choice of eccentricity. Assume that $M_1 = 0.5\ \mathrm{M}_\odot$, $M_2 = 2.0\ \mathrm{M}_\odot$, $P = 1.8$ yr, and $i = 30°$. Plot your results for $e = 0, 0.2, 0.4,$ and 0.5. (You may assume that the center-of-mass velocity is zero and that the orientation of the major axis is perpendicular to the line of sight.)

Assume that the radii and effective temperatures of the two stars are $R_1 = 1.8\ \mathrm{R}_\odot$, $T_{e1} = 8190$ K, and $R_2 = 0.63\,\mathrm{R}_\odot$, $T_{e2} = 3840$ K.

(b) Verify your results for $e = 0$ by using the equations developed in Section 7.3.

(c) Explain how you might determine the eccentricity of an orbital system.

7.16 The code `TwoStars` (Appendix K) can be used to analyze the apparent motions of binary stars across the plane of the sky. If fact, `TwoStars` was used to generate the data for Fig. 7.1. Assume that the binary system used in Problem 7.15 is located 3.2 pc from Earth and that its center of mass is moving through space with the vector components $(v_x', v_y', v_z') = (30\ \mathrm{km\ s^{-1}}, 42\ \mathrm{km\ s^{-1}}, -15.3\ \mathrm{km\ s^{-1}})$. From the position data generated by `TwoStars`, plot the apparent positions of the stars in milliarcseconds for the case where $e = 0.4$.

7.17 Figure 7.2 shows the light curve of the eclipsing binary YY Sgr. The code `TwoStars`, described in Appendix K and available on the companion website, can be used to roughly model this system. Use the data provided in the caption, and assume that the masses, radii, and effective temperatures of the two stars are $M_1 = 5.9\ \mathrm{M}_\odot$, $R_1 = 3.2\ \mathrm{R}_\odot$, $T_{e1} = 15{,}200$ K, and $M_2 = 5.6\ \mathrm{M}_\odot$, $R_2 = 2.9\ \mathrm{R}_\odot$, $T_{e2} = 13{,}700$ K. Also assume that the periastron angle is 214.6° and that the center of mass is at rest relative to the observer.

(a) Using `TwoStars`, create a synthetic light curve for the system.

(b) Using `TwoStars`, plot the radial velocities of the two stars.

7.18 Using the data given in the text, and assuming that the orbital inclination is 90°, use `TwoStars` (Appendix K) to generate data that model the light curve of OGLE-TR-56b. You may assume that the radius of the planet is approximately the radius of Jupiter (7×10^7 m) and its temperature is roughly 1000 K. Take the temperature of the star to be 3000 K. You may also assume that the planet's orbit is perfectly circular.

CHAPTER

8 The Classification of Stellar Spectra

8.1 ■ THE FORMATION OF SPECTRAL LINES

With the invention of photometry and spectroscopy, the new science of *astrophysics* progressed rapidly. As early as 1817, Joseph Fraunhofer had determined that different stars have different spectra. Stellar spectra were classified according to several schemes, the earliest of which recognized just three types of spectra. As instruments improved, increasingly subtle distinctions became possible.

The Spectral Types of Stars

A spectral taxonomy developed at Harvard by Edward C. Pickering (1846–1919) and his assistant Williamina P. Fleming (1857–1911) in the 1890s labeled spectra with capital letters according to the strength of their hydrogen absorption lines, beginning with the letter A for the broadest lines. At about the same time, Antonia Maury (1866–1952), another of Pickering's assistants and a colleague of Fleming's, was developing a somewhat different classification scheme that she was using to study the widths of spectral lines. In her work Maury rearranged her classes in a way that would have been equivalent to placing Pickering's and Fleming's B class before the A stars. Then, in 1901, Annie Jump Cannon[1] (1863–1941; see Fig. 8.1), also employed by Pickering, and using the scheme of Pickering and Fleming while following the suggestion of Maury, rearranged the sequence of spectra by placing O and B before A, added decimal subdivisions (e.g., A0–A9), and consolidated many of the classes. With these changes, the Harvard classification scheme of "O B A F G K M" became a *temperature* sequence, running from the hottest blue O stars to the coolest red M stars. Generations of astronomy students have remembered this string of **spectral types** by memorizing the phrase "Oh Be A Fine Girl/Guy, Kiss Me." Stars nearer the beginning of this sequence are referred to as **early-type** stars, and those closer to the end are called **late-type** stars. These labels also distinguish the stars within the spectral subdivisions, so astronomers may speak of a K0 star as an "early K star" or refer to a B9 star as a "late B star." Cannon classified some 200,000 spectra between 1911 and 1914, and the results were

[1]The Annie J. Cannon Award is bestowed annually by the American Association of University Women and the American Astronomical Society for distinguished contributions to astronomy by a woman.

FIGURE 8.1 Annie Jump Cannon (1863–1941). (Courtesy of Harvard College Observatory.)

collected into the **Henry Draper Catalogue**.[2] Today, many stars are referred to by their HD numbers; Betelgeuse is HD 39801.

The physical basis of the Harvard spectral classification scheme remained obscure, however. Vega (spectral type A0) displays very strong hydrogen absorption lines, much stronger than the faint lines observed for the Sun (spectral type G2). On the other hand, the Sun's calcium absorption lines are much more intense than those of Vega. Is this a result of a variation in the *composition* of the two stars? Or are the different surface temperatures of Vega ($T_e = 9500$ K) and the Sun ($T_e = 5777$ K) responsible for the relative strengths of the absorption lines?

The theoretical understanding of the quantum atom achieved early in the twentieth century gave astronomers the key to the secrets of stellar spectra. As discussed in Section 5.3, absorption lines are created when an atom absorbs a photon with exactly the energy required for an electron to make an upward transition from a lower to a higher orbital. Emission lines are formed in the inverse process, when an electron makes a downward transition from a higher to a lower orbital and a single photon carries away the energy lost by the electron. The wavelength of the photon thus depends on the energies of the atomic orbitals involved in these transitions. For example, the Balmer absorption lines of hydrogen are caused by electrons making upward transitions from the $n = 2$ orbital to higher-energy orbitals, and Balmer emission lines are produced when electrons make downward transitions from higher-energy orbitals to the $n = 2$ orbital.

The distinctions between the spectra of stars with different temperatures are due to electrons occupying different atomic orbitals in the atmospheres of these stars. The details of spectral line formation can be quite complicated because electrons can be found in any of an atom's orbitals. Furthermore, the atom can be in any one of various stages of ionization and has a unique set of orbitals at each stage. An atom's stage of ionization is denoted by a

[2]In 1872 Henry Draper took the first photograph of a stellar spectrum. The catalog bearing his name was financed from his estate.

Roman numeral following the symbol for the atom. For example, H I and He I are neutral (not ionized) hydrogen and helium, respectively; He II is singly ionized helium, and Si III and Si IV refer to a silicon atom that has lost two and three electrons, respectively.

In the Harvard system devised by Cannon, the Balmer lines reach their maximum intensity in the spectra of stars of type A0, which have an effective temperature of $T_e = 9520$ K (recall Eq. 3.17). The visible spectral lines of neutral helium (He I) are strongest for B2 stars ($T_e = 22{,}000$ K), and the visible spectral lines of singly ionized calcium (Ca II) are most intense for K0 stars ($T_e = 5250$ K).[3]

Table 8.1 lists some of the defining criteria for various spectral types. In the table the term **metal** is used to indicate any element heavier than helium, a convention commonly adopted by astronomers because by far the most abundant elements in the universe are hydrogen and helium.

In addition to the traditional spectral types of the Harvard classification scheme (OBAFGKM), Table 8.1 also includes recently defined spectral types of very cool stars and **brown dwarfs**. Brown dwarfs are objects with too little mass to allow nuclear reactions to occur in their interiors in any substantial way, so they are not considered stars in the usual sense (see Section 10.6). The necessity of introducing these new spectral types came from all-sky surveys that detected a large number of objects with very low effective temperatures (1300 K to 2500 K for L spectral types and less than 1300 K for T spectral types).[4] In order to remember the new, cooler spectral types, one might consider extending the popular mnemonic by: "Oh Be A Fine Girl/Guy, Kiss Me—Less Talk!"

Figures 8.2 and 8.3 display some sample photographic spectra for various spectral types. You will note that hydrogen lines [e.g., Hγ (434.4 nm) and Hδ (410.1 nm)] increase in width (strength) from O9 to A0, then decrease in width from A0 through F5, and nearly vanish by late K. Helium (He) lines are discernible in the spectra of early-type stars (O and early B) but begin to disappear in cooler stars.

Figures 8.4 and 8.5 also depict stellar spectra in a graphical format typical of modern digital detectors. Readily apparent is the shifting to longer wavelengths of the peak of the superimposed blackbody spectrum as the temperature of the star decreases (later spectral types). Also apparent are the Hα, Hβ, Hγ, and Hδ Balmer lines at 656.2 nm, 486.1 nm, 434.0 nm, and 410.2 nm, respectively. Note how these hydrogen absorption lines grow in strength from O to A and then decrease in strength for spectral types later than A. For later spectral types, the messy spectra are indicative of metal lines, with molecular lines appearing in the spectra of the coolest stars.

The Maxwell–Boltzmann Velocity Distribution

To uncover the physical foundation of this classification system, two basic questions must be answered: In what orbitals are electrons most likely to be found? What are the relative numbers of atoms in various stages of ionization?

[3]The two prominent spectral lines of Ca II are usually referred to as the H ($\lambda = 396.8$ nm) and K ($\lambda = 393.3$ nm) lines of calcium. The nomenclature for the H line was devised by Fraunhofer; the K line was named by E. Mascart (1837–1908) in the 1860s.

[4]The surveys that discovered large numbers of these objects are the Sloan Digital Sky Survey (SDSS) and the 2-Micron All-Sky Survey (2MASS).

TABLE 8.1 Harvard Spectral Classification.

Spectral Type	Characteristics
O	Hottest blue-white stars with few lines Strong He II absorption (sometimes emission) lines. He I absorption lines becoming stronger.
B	Hot blue-white He I absorption lines strongest at B2. H I (Balmer) absorption lines becoming stronger.
A	White Balmer absorption lines strongest at A0, becoming weaker later. Ca II absorption lines becoming stronger.
F	Yellow-white Ca II lines continue to strengthen as Balmer lines continue to weaken. Neutral metal absorption lines (Fe I, Cr I).
G	Yellow Solar-type spectra. Ca II lines continue becoming stronger. Fe I, other neutral metal lines becoming stronger.
K	Cool orange Ca II H and K lines strongest at K0, becoming weaker later. Spectra dominated by metal absorption lines.
M	Cool red Spectra dominated by molecular absorption bands, especially titanium oxide (TiO) and vanadium oxide (VO). Neutral metal absorption lines remain strong.
L	Very cool, dark red Stronger in infrared than visible. Strong molecular absorption bands of metal hydrides (CrH, FeH), water (H_2O), carbon monoxide (CO), and alkali metals (Na, K, Rb, Cs). TiO and VO are weakening.
T	Coolest, Infrared Strong methane (CH_4) bands but weakening CO bands.

S and C spectral types for evolved giant stars are discussed on page 466.

The answers to both questions are found in an area of physics known as **statistical mechanics**. This branch of physics studies the statistical properties of a system composed of many members. For example, a gas can contain a huge number of particles with a large range of speeds and energies. Although in practice it would be impossible to calculate the detailed behavior of any single particle, the gas as a whole does have certain well-defined properties, such as its temperature, pressure, and density. For such a gas in thermal

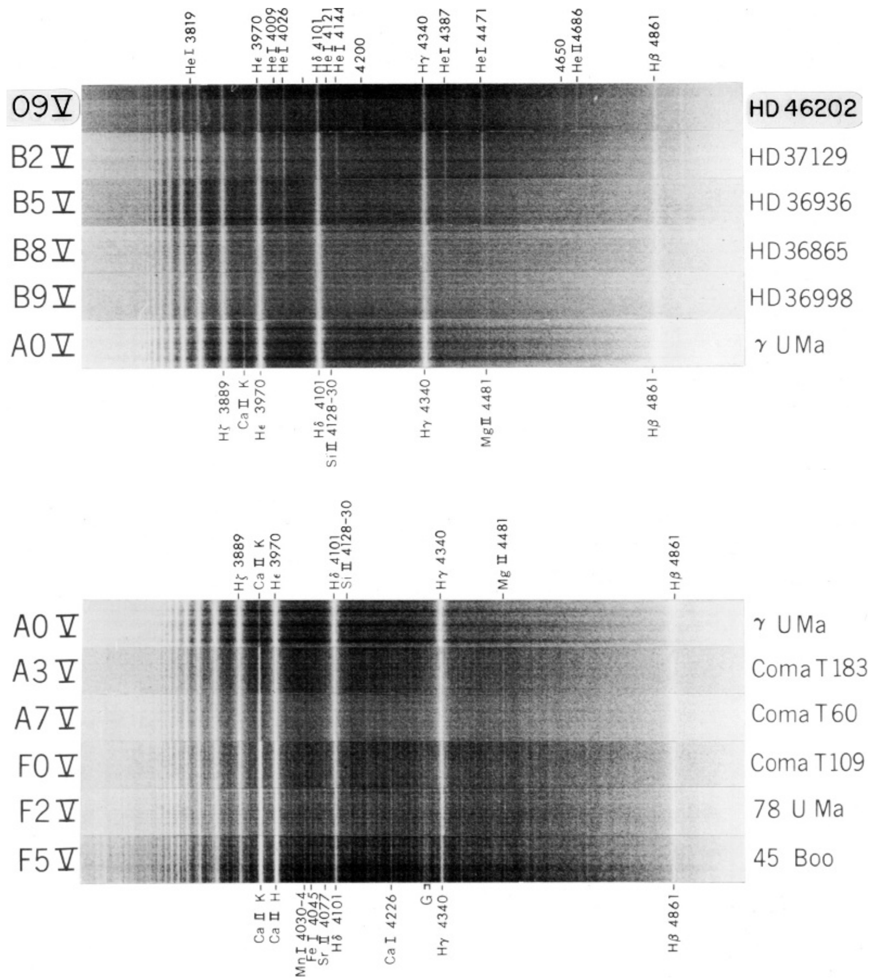

FIGURE 8.2 Stellar spectra for main-sequence classes O9–F5. Note that these spectra are displayed as negatives; absorption lines appear bright. Wavelengths are given in angstroms. (Figure from Abt, et al., *An Atlas of Low-Dispersion Grating Stellar Spectra*, Kitt Peak National Observatory, Tucson, AZ, 1968.)

equilibrium (the gas is not rapidly increasing or decreasing in temperature, for instance), the **Maxwell–Boltzmann velocity distribution function**[5] describes the fraction of particles having a given range of speeds. The number of gas particles per unit volume having speeds between v and $v + dv$ is given by

$$\boxed{n_v\,dv = n\left(\frac{m}{2\pi kT}\right)^{3/2} e^{-mv^2/2kT}\,4\pi v^2\,dv,} \tag{8.1}$$

[5]This name honors James Clerk Maxwell and Ludwig Boltzmann (1844–1906), the latter of whom is considered the founder of statistical mechanics.

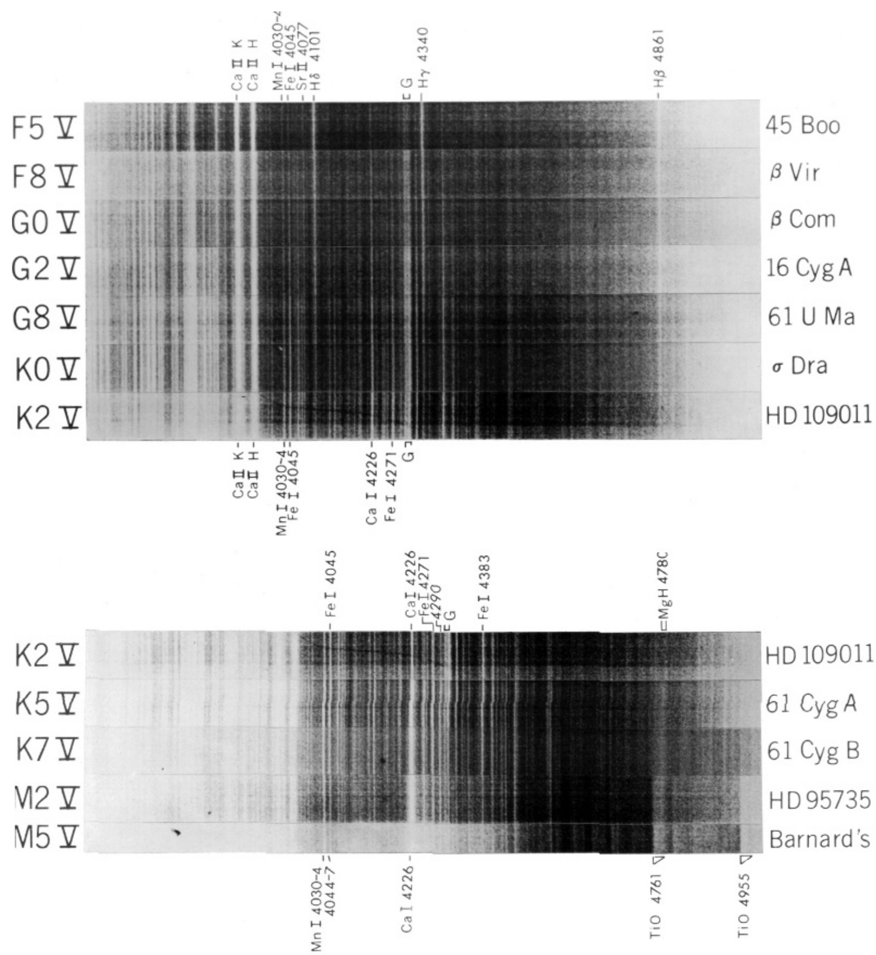

FIGURE 8.3 Stellar spectra for main-sequence classes F5–M5. Note that these spectra are displayed as negatives; absorption lines appear bright. Wavelengths are given in angstroms. (Figure from Abt, et al., *An Atlas of Low-Dispersion Grating Stellar Spectra*, Kitt Peak National Observatory, Tucson, AZ, 1968.)

where n is the total number density (number of particles per unit volume), $n_v \equiv \partial n/\partial v$, m is a particle's mass, k is Boltzmann's constant, and T is the temperature of the gas *in kelvins*. Figure 8.6 shows the Maxwell–Boltzmann distribution of molecular speeds in terms of the *fraction* of molecules having a speed between v and $v + dv$. The exponent of the distribution function is the ratio of a gas particle's kinetic energy, $\frac{1}{2}mv^2$, to the characteristic thermal energy, kT. It is difficult for a significant number of particles to have an energy much greater or less than the thermal energy; the distribution peaks when these energies are equal, at a **most probable speed** of

$$v_{\rm mp} = \sqrt{\frac{2kT}{m}}. \tag{8.2}$$

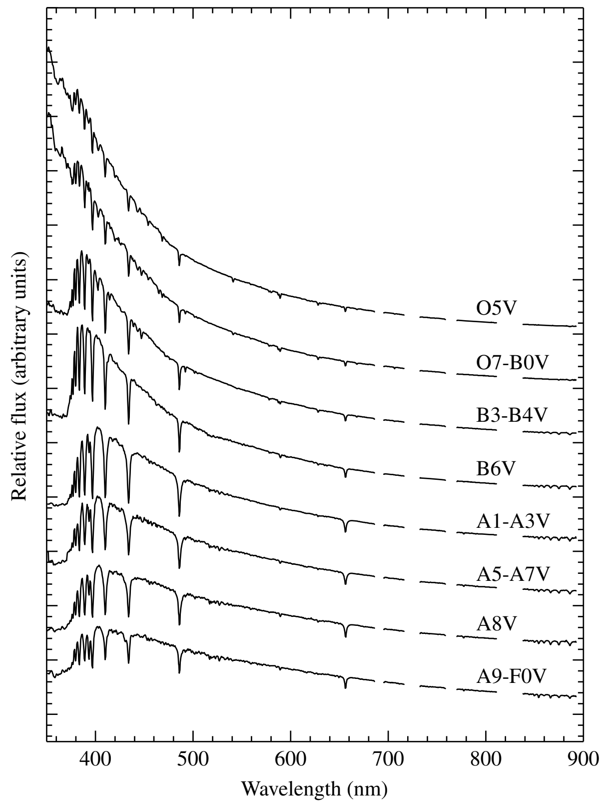

FIGURE 8.4 Digitized spectra of main sequence classes O5–F0 displayed in terms of relative flux as a function of wavelength. Modern spectra obtained by digital detectors (as opposed to photographic plates) are generally displayed graphically. (Data from Silva and Cornell, *Ap. J. Suppl.*, *81*, 865, 1992.)

The high-speed exponential "tail" of the distribution function results in a somewhat higher (average) **root-mean-square speed**[6] of

$$\boxed{v_{\rm rms} = \sqrt{\frac{3kT}{m}}.} \tag{8.3}$$

Example 8.1.1. The area under the curve between two speeds is equal to the fraction of gas particles in that range of speeds. In order to determine the fraction of hydrogen atoms in a gas of $T = 10{,}000$ K having speeds between $v_1 = 2 \times 10^4$ m s^{-1} and $v_2 = 2.5 \times 10^4$ m s^{-1}, it is necessary to integrate the Maxwell–Boltzmann distribution between these two limits,

[6]The root-mean-square speed is the square root of the average (mean) value of v^2: $v_{\rm rms} = \sqrt{\overline{v^2}}$.

or

$$\begin{aligned} N/N_{\text{total}} &= \frac{1}{n}\int_{v_1}^{v_2} n_v\,dv \\ &= \left(\frac{m}{2\pi kT}\right)^{3/2}\int_{v_1}^{v_2} e^{-mv^2/2kT}\,4\pi v^2\,dv. \end{aligned} \tag{8.4}$$

Although Eq. (8.4) has a closed-form solution when $v_1 = 0$ and $v_2 \to \infty$, it must be evaluated numerically in other cases. This can be accomplished crudely by evaluating the integrand using an average value of the velocity over the interval, multiplied by the width of the interval, or

$$N/N_{\text{total}} = \frac{1}{n}\int_{v_1}^{v_2} n_v(v)\,dv \simeq \frac{1}{n}n_v(\overline{v})\,(v_2 - v_1),$$

where $\overline{v} \equiv (v_1 + v_2)/2$. Substituting, we find

$$\begin{aligned} N/N_{\text{total}} &\simeq \left(\frac{m}{2\pi kT}\right)^{3/2} e^{-m\overline{v}^2/2kT}\,4\pi\overline{v}^2\,(v_2 - v_1) \\ &\simeq 0.125. \end{aligned}$$

Approximately 12.5% of the hydrogen atoms in a gas at 10,000 K have speeds between 2×10^4 m s^{-1} and 2.5×10^4 m s^{-1}. A more careful numerical integration over the range gives 12.76%.

The Boltzmann Equation

The atoms of a gas gain and lose energy as they collide. As a result, the distribution in the speeds of the impacting atoms, given by Eq. (8.1), produces a definite distribution of the electrons among the atomic orbitals. This distribution of electrons is governed by a fundamental result of statistical mechanics: Orbitals of higher energy are less likely to be occupied by electrons.

Let s_a stand for the specific set of quantum numbers that identifies a state of energy E_a for a system of particles. Similarly, let s_b stand for the set of quantum numbers that identifies a state of energy E_b. For example, $E_a = -13.6$ eV for the lowest orbit of the hydrogen atom, with $s_a = \{n = 1,\ \ell = 0,\ m_\ell = 0,\ m_s = +1/2\}$ identifying a specific state with that energy (recall Section 5.4 for a discussion of quantum numbers). Then the ratio of the probability $P(s_b)$ that the system is in state s_b to the probability $P(s_a)$ that the system is in state s_a is given by

$$\frac{P(s_b)}{P(s_a)} = \frac{e^{-E_b/kT}}{e^{-E_a/kT}} = e^{-(E_b - E_a)/kT}, \tag{8.5}$$

where T is the common temperature of the two systems. The term $e^{-E/kT}$ is called the *Boltzmann factor*.[7]

[7]The energies encountered in this context are usually given in units of electron volts (eV), so it is useful to remember that at a room temperature of 300 K, the product kT is approximately 1/40 eV.

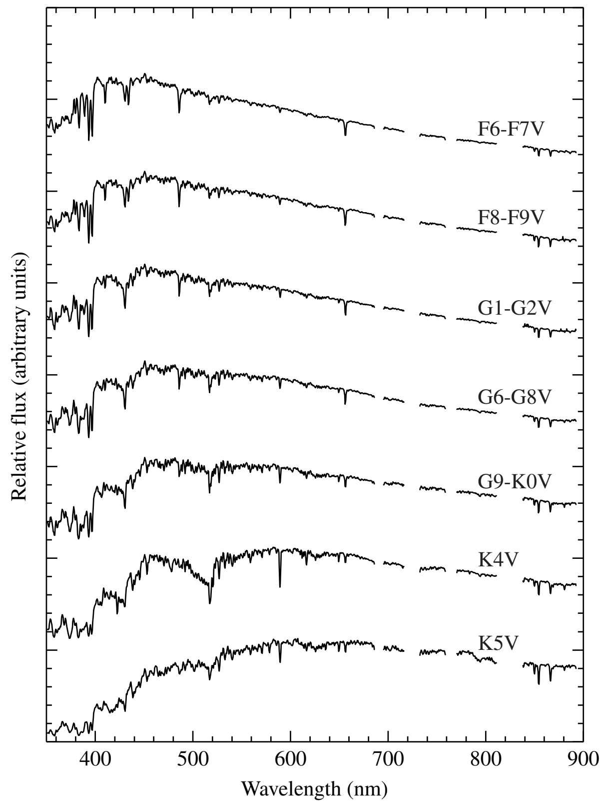

FIGURE 8.5 Digitized spectra of main sequence classes F6–K5 displayed in terms of relative flux as a function of wavelength. (Data from Silva and Cornell, *Ap. J. Suppl.*, *81*, 865, 1992.)

The Boltzmann factor plays such a fundamental role in the study of statistical mechanics that Eq. (8.5) merits further consideration. Suppose, for example, that $E_b > E_a$; the energy of state s_b is greater than the energy of state s_a. Notice that as the thermal energy kT decreases toward zero (i.e., $T \to 0$), the quantity $-(E_b - E_a)/kT \to -\infty$, and so $P(s_b)/P(s_a) \to 0$. This is just what is to be expected if there isn't any thermal energy available to raise the energy of an atom to a higher level. On the other hand, if there is a great deal of thermal energy available (i.e., $T \to \infty$), then $-(E_b - E_a)/kT \to 0$ and $P(s_b)/P(s_a) \to 1$. Again this is what would be expected since with an unlimited reservoir of thermal energy, all available energy levels of the atom should be accessible with equal probability. You can quickly verify that if we had assumed instead that $E_b < E_a$, the expected results would again be obtained in the limits of $T \to 0$ and $T \to \infty$.

It is often the case that the energy levels of the system may be **degenerate**, with more than one quantum state having the same energy. That is, if states s_a and s_b are degenerate, then $E_a = E_b$ but $s_a \neq s_b$. When taking averages, we must count each of the degenerate states separately. To account properly for the number of states that have a given energy, define g_a to be the number of states with energy E_a. Similarly, define g_b to be the number of states with energy E_b. These are called the **statistical weights** of the energy levels.

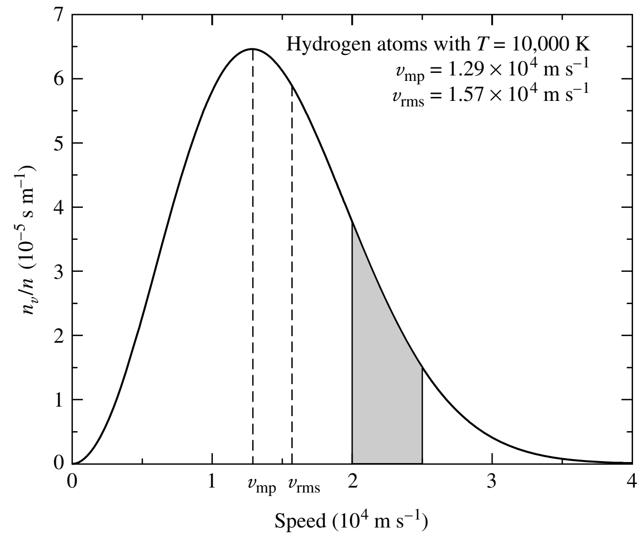

FIGURE 8.6 Maxwell–Boltzmann distribution function, n_v/n, for hydrogen atoms at a temperature of 10,000 K. The fraction of hydrogen atoms in the gas having velocities between 2×10^4 m s^{-1} and 2.5×10^4 m s^{-1} is the shaded area under the curve between those two velocities; see Example 8.1.1.

Example 8.1.2. The ground state of the hydrogen atom is twofold degenerate. In fact, although "ground state" is the standard terminology, the plural "ground state*s*" would be more precise because these are *two* quantum states that have the same energy of -13.6 eV (for $m_s = \pm 1/2$).[8] In the same manner, the "first excited state" actually consists of *eight* degenerate quantum states with the same energy of -3.40 eV.

Table 8.2 shows the set of quantum numbers $\{n, \ell, m_\ell, m_s\}$ that identifies each state; it also shows each state's energy. Notice that there are $g_1 = 2$ ground states with the energy $E_1 = -13.6$ eV, and $g_2 = 8$ first excited states with the energy $E_2 = -3.40$ eV. This result agrees with that of Problem 5.16, which demonstrated that the degeneracy of energy level n of the hydrogen atom is $2n^2$.

The ratio of the probability $P(E_b)$ that the system will be found in *any* of the g_b degenerate states with energy E_b to the probability $P(E_a)$ that the system is in *any* of the g_a degenerate states with energy E_a is given by

$$\frac{P(E_b)}{P(E_a)} = \frac{g_b\, e^{-E_b/kT}}{g_a\, e^{-E_a/kT}} = \frac{g_b}{g_a}\, e^{-(E_b-E_a)/kT}.$$

Stellar atmospheres contain a vast number of atoms, so the ratio of probabilities is indistinguishable from the ratio of the number of atoms. Thus, for the atoms of a given element in a specified state of ionization, the ratio of the number of atoms N_b with energy E_b to

[8]In reality, the two "ground states" of the hydrogen atom are not precisely degenerate. As explained in Section 12.1, the two states actually have slightly different energies, enabling the hydrogen atom to emit 21-cm radio waves, an important signature of hydrogen gas in interstellar space.

TABLE 8.2 Quantum Numbers and Energies for the Hydrogen Atom.

Ground States s_1				Energy E_1
n	ℓ	m_ℓ	m_s	(eV)
1	0	0	+1/2	−13.6
1	0	0	−1/2	−13.6

First Excited States s_2				Energy E_2
n	ℓ	m_ℓ	m_s	(eV)
2	0	0	+1/2	−3.40
2	0	0	−1/2	−3.40
2	1	1	+1/2	−3.40
2	1	1	−1/2	−3.40
2	1	0	+1/2	−3.40
2	1	0	−1/2	−3.40
2	1	−1	+1/2	−3.40
2	1	−1	−1/2	−3.40

the number of atoms N_a with energy E_a *in different states of excitation* is given by the **Boltzmann equation**,

$$\boxed{\frac{N_b}{N_a} = \frac{g_b\, e^{-E_b/kT}}{g_a\, e^{-E_a/kT}} = \frac{g_b}{g_a}\, e^{-(E_b-E_a)/kT}.} \tag{8.6}$$

Example 8.1.3. For a gas of neutral hydrogen atoms, at what temperature will equal numbers of atoms have electrons in the ground state ($n = 1$) and in the first excited state ($n = 2$)?[9] Recall from Example 8.1.2 that the degeneracy of the nth energy level of the hydrogen atom is $g_n = 2n^2$. Associating state a with the ground state and state b with the first excited state, setting $N_2 = N_1$ on the left-hand side of Eq. (8.6), and using Eq. (5.14) for the energy levels lead to

$$1 = \frac{2(2)^2}{2(1)^2}\, e^{-[(-13.6 \text{ eV}/2^2)-(-13.6 \text{ eV}/1^2)]/kT},$$

or

$$\frac{10.2 \text{ eV}}{kT} = \ln{(4)}.$$

[9]We have reverted to the standard practice of referring to the two degenerate states of lowest energy as the "ground state" and to the eight degenerate states of next-lowest energy as the "first excited state."

Solving for the temperature yields[10]

$$T = \frac{10.2 \text{ eV}}{k \ln(4)} = 8.54 \times 10^4 \text{ K}.$$

High temperatures are required for a significant number of hydrogen atoms to have electrons in the first excited state. Figure 8.7 shows the relative occupancy of the ground and first excited states, $N_2/(N_1 + N_2)$, as a function of temperature.[11] This result is somewhat puzzling, however. Recall that the Balmer absorption lines are produced by electrons in hydrogen atoms making an upward transition from the $n = 2$ orbital. If, as shown in Example 8.1.3, temperatures on the order of 85,000 K are needed to provide electrons in the first excited state, then why do the Balmer lines reach their maximum intensity at a much lower temperature of 9520 K? Clearly, according to Eq. (8.6), at temperatures higher than 9520 K an even greater proportion of the electrons will be in the first excited state rather than in the ground state. If this is the case, then what is responsible for the diminishing strength of the Balmer lines at higher temperatures?

The Saha Equation

The answer lies in also considering the relative number of atoms *in different stages of ionization*. Let χ_i be the ionization energy needed to remove an electron from an atom (or

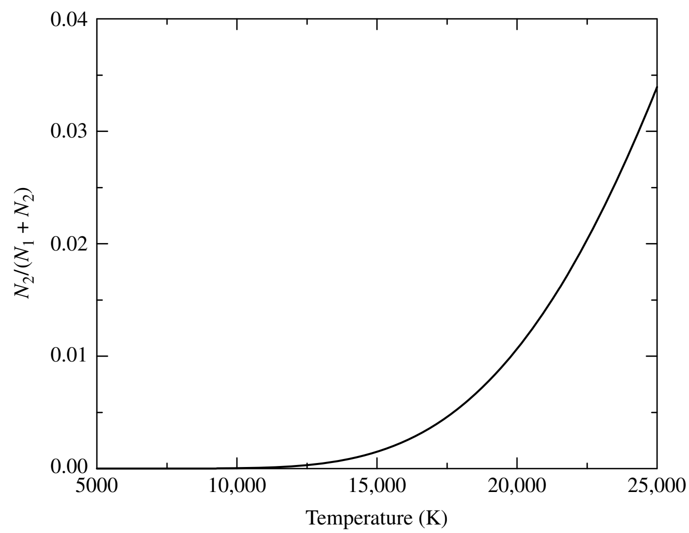

FIGURE 8.7 $N_2/(N_1 + N_2)$ for the hydrogen atom obtained via the Boltzmann equation.

[10]When we are working with electron volts, the Boltzmann constant can be expressed in the convenient form $k = 8.6173423 \times 10^{-5}$ eV K^{-1}.

[11]For the remainder of this section, we will use $a = 1$ for the ground state energy and $b = 2$ for the energy of the first excited state.

ion) in the ground state, thus taking it from ionization stage i to stage $(i + 1)$. For example, the ionization energy of hydrogen, the energy needed to convert it from H I to H II, is $\chi_I = 13.6$ eV. However, it may be that the initial and final ions are not in the ground state. An average must be taken over the orbital energies to allow for the possible partitioning of the atom's electrons among its orbitals. This procedure involves calculating the **partition functions**, Z, for the initial and final atoms. The partition function is simply the weighted sum of the number of ways the atom can arrange its electrons with the same energy, with more energetic (and therefore less likely) configurations receiving less weight from the Boltzmann factor when the sum is taken. If E_j is the energy of the jth energy level and g_j is the degeneracy of that level, then the partition function Z is defined as

$$\boxed{Z = \sum_{j=1}^{\infty} g_j\, e^{-(E_j - E_1)/kT}.} \tag{8.7}$$

If we use the partition functions Z_i and Z_{i+1} for the atom in its initial and final stages of ionization, the ratio of the number of atoms in stage $(i + 1)$ to the number of atoms in stage i is

$$\boxed{\frac{N_{i+1}}{N_i} = \frac{2Z_{i+1}}{n_e Z_i} \left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_i/kT}.} \tag{8.8}$$

This equation is known as the **Saha equation**, after the Indian astrophysicist Meghnad Saha (1894–1956), who first derived it in 1920. Because a free electron is produced in the ionization process, it is not surprising to find the number density of free electrons (number of free electrons per unit volume), n_e, on the right-hand side of the Saha equation. Note that as the number density of free electrons increases, the number of atoms in the higher stage of ionization decreases, since there are more electrons with which the ion may recombine. The factor of 2 in front of the partition function Z_{i+1} reflects the two possible spins of the free electron, with $m_s = \pm 1/2$. The term in parentheses is also related to the free electron, with m_e being the electron mass.[12] Sometimes the pressure of the free electrons, P_e, is used in place of the electron number density; the two are related by the ideal gas law written in the form

$$P_e = n_e kT.$$

Then the Saha equation takes the alternative form

$$\boxed{\frac{N_{i+1}}{N_i} = \frac{2kT Z_{i+1}}{P_e Z_i} \left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_i/kT}.} \tag{8.9}$$

[12]The term in parentheses is the number density of electrons for which the quantum energy (such as discussed in Example 5.4.2) is roughly equal to the characteristic thermal energy kT. For the classical conditions encountered in stellar atmospheres, this term is much greater than n_e.

The electron pressure ranges from 0.1 N m^{-2} in the atmospheres of cooler stars to 100 N m^{-2} for hotter stars. In Section 9.5, we will describe how the electron pressure is determined for stellar atmospheres.

Combining the Boltzmann and Saha Equations

We are now finally ready to consider the combined effects of the Boltzmann and Saha equations and how they influence the stellar spectra that we observe.

Example 8.1.4. Consider the degree of ionization in a stellar atmosphere that is assumed to be composed of pure hydrogen. Assume for simplicity that the electron pressure is a constant $P_e = 20$ N m^{-2}.

The Saha equation (8.9) will be used to calculate the fraction of atoms that are ionized, $N_{\rm II}/N_{\rm total} = N_{\rm II}/(N_{\rm I} + N_{\rm II})$, as the temperature T varies between 5000 K and 25,000 K. However, the partition functions $Z_{\rm I}$ and $Z_{\rm II}$ must be determined first. A hydrogen ion is just a proton and so has no degeneracy; thus $Z_{\rm II} = 1$. The energy of the first excited state of hydrogen is $E_2 - E_1 = 10.2$ eV above the ground state energy. Because 10.2 eV $\gg kT$ for the temperature regime under consideration, the Boltzmann factor $e^{-(E_2-E_1)/kT} \ll 1$. Nearly all of the H I atoms are therefore in the ground state (recall the previous example), so Eq. (8.7) for the partition function simplifies to $Z_{\rm I} \simeq g_1 = 2(1)^2 = 2$.

Inserting these values into the Saha equation with $\chi_{\rm I} = 13.6$ eV gives the ratio of ionized to neutral hydrogen, $N_{\rm II}/N_{\rm I}$. This ratio is then used to find the fraction of ionized hydrogen, $N_{\rm II}/N_{\rm total}$, by writing

$$\frac{N_{\rm II}}{N_{\rm total}} = \frac{N_{\rm II}}{N_{\rm I} + N_{\rm II}} = \frac{N_{\rm II}/N_{\rm I}}{1 + N_{\rm II}/N_{\rm I}};$$

the results are displayed in Fig. 8.8. This figure shows that when $T = 5000$ K, essentially none of the hydrogen atoms are ionized. At about 8300 K, 5% of the atoms have become ionized. Half of the hydrogen is ionized at a temperature of 9600 K, and when T has risen to 11,300 K, all but 5% of the hydrogen is in the form of H II. Thus the ionization of hydrogen takes place within a temperature interval of approximately 3000 K. This range of temperatures is quite limited compared to the temperatures of tens of millions of degrees routinely encountered inside stars. The narrow region inside a star where hydrogen is partially ionized is called a hydrogen **partial ionization zone** and has a characteristic temperature of approximately 10,000 K for a wide range of stellar parameters.

Now we can see why the Balmer lines are observed to attain their maximum intensity at a temperature of 9520 K, instead of at the much higher characteristic temperatures (on the order of 85,000 K) required to excite electrons to the $n = 2$ energy level of hydrogen. The strength of the Balmer lines depends on $N_2/N_{\rm total}$, the fraction of *all* hydrogen atoms that are in the first excited state. This is found by combining the results of the Boltzmann and Saha equations. Because virtually all of the neutral hydrogen atoms are in either the ground state or the first excited state, we can employ the approximation $N_1 + N_2 \simeq N_{\rm I}$ and write

$$\frac{N_2}{N_{\rm total}} = \left(\frac{N_2}{N_1 + N_2}\right)\left(\frac{N_{\rm I}}{N_{\rm total}}\right) = \left(\frac{N_2/N_1}{1 + N_2/N_1}\right)\left(\frac{1}{1 + N_{\rm II}/N_{\rm I}}\right).$$

continued

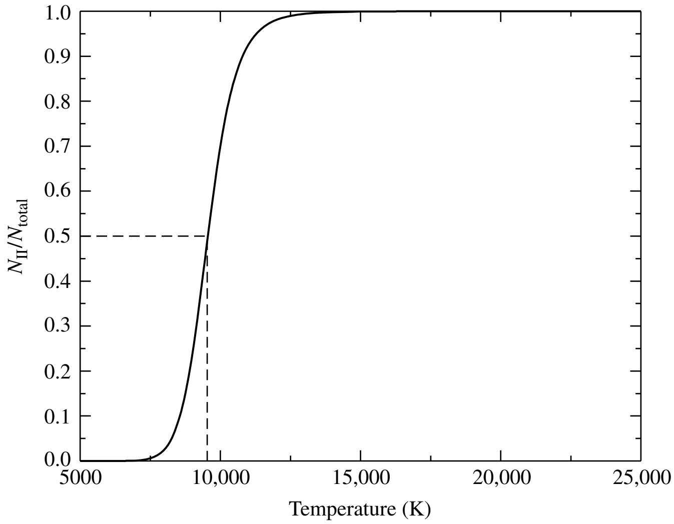

FIGURE 8.8 $N_{\mathrm{II}}/N_{\mathrm{total}}$ for hydrogen from the Saha equation when $P_e = 20\ \mathrm{N\ m^{-2}}$. Fifty percent ionization occurs at $T \simeq 9600$ K.

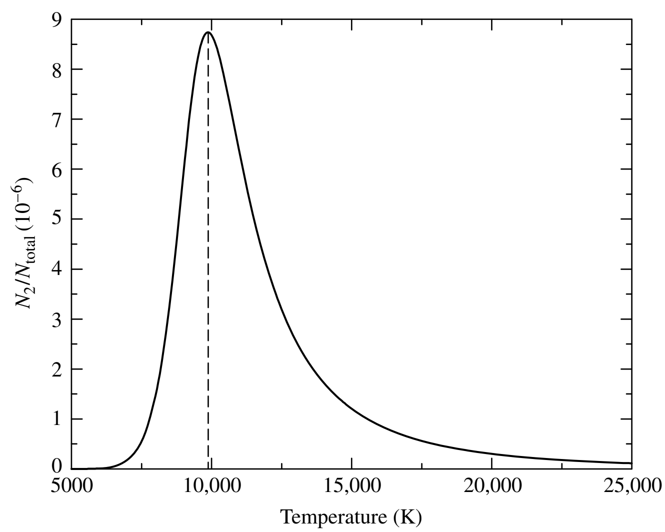

FIGURE 8.9 N_2/N_{total} for hydrogen from the Boltzmann and Saha equations, assuming $P_e = 20\ \mathrm{N\ m^{-2}}$. The peak occurs at approximately 9900 K.

Figure 8.9 shows that in this example, the hydrogen gas would produce the most intense Balmer lines at a temperature of 9900 K, in good agreement with the observations. *The diminishing strength of the Balmer lines at higher temperatures is due to the rapid ionization of hydrogen above 10,000 K.* Figure 8.10 summarizes this situation.

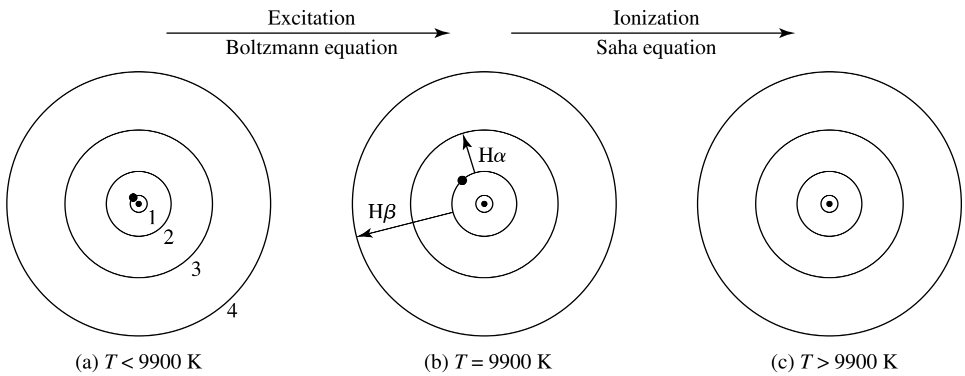

FIGURE 8.10 The electron's position in the hydrogen atom at different temperatures. In (a), the electron is in the ground state. Balmer absorption lines are produced only when the electron is initially in the first excited state, as shown in (b). In (c), the atom has been ionized.

Of course, stellar atmospheres are not composed of pure hydrogen, and the results obtained in Example 8.1.4 depended on an appropriate value for the electron pressure. In stellar atmospheres, there is typically one helium atom for every ten hydrogen atoms. The presence of ionized helium provides more electrons with which the hydrogen ions can recombine. Thus, when helium is added, it takes a *higher* temperature to achieve the same degree of hydrogen ionization.

It should also be emphasized that the Saha equation can be applied only to a gas in *thermodynamic equilibrium*, so that the Maxwell–Boltzmann velocity distribution is obeyed.[13] Furthermore, the density of the gas must not be too great (less than roughly 1 kg m^{-3} for stellar material), or the presence of neighboring ions will distort an atom's orbitals and lower its ionization energy.

Example 8.1.5. The Sun's "surface" is a thin layer of the solar atmosphere called the *photosphere*; see Section 11.2. The characteristic temperature of the photosphere is $T = T_e = 5777$ K, and it has about 500,000 hydrogen atoms for each calcium atom with an electron pressure of about 1.5 N m^{-2}.[14] From this information and knowledge of the appropriate statistical weights and partition functions, the Saha and Boltzmann equations can be used to estimate the relative strengths of the absorption lines due to hydrogen (the Balmer lines) and those due to calcium (the Ca II H and K lines).

We must compare the number of neutral hydrogen atoms with electrons in the first excited state (which produce the Balmer lines) to the number of singly ionized calcium atoms with electrons in the ground state (which produce the Ca II H and K lines). As in Example 8.1.4, we will use the Saha equation to determine the degree of ionization and will use the Boltzmann equation to reveal the distribution of electrons between the ground and first excited states.

continued

[13]Thermodynamic equilibrium will be discussed in detail in Section 9.2.
[14]See Cox (2000), page 348 for a model solar photosphere.

Let's consider hydrogen first. If we substitute the partition functions found in Example 8.1.4 into the Saha equation (8.9), the ratio of ionized to neutral hydrogen is

$$\left[\frac{N_{\mathrm{II}}}{N_{\mathrm{I}}}\right]_{\mathrm{H}} = \frac{2kTZ_{i+1}}{P_e Z_i}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_i/kT} = 7.70\times 10^{-5} \simeq \frac{1}{13{,}000}.$$

Thus there is only one hydrogen ion (H II) for every 13,000 neutral hydrogen atoms (H I) at the Sun's surface. Almost none of the hydrogen is ionized.

The Boltzmann equation (8.6) reveals how many of these neutral hydrogen atoms are in the first excited state. Using $g_n = 2n^2$ for hydrogen (implying $g_1 = 2$ and $g_2 = 8$), we have

$$\left[\frac{N_2}{N_1}\right]_{\mathrm{H\ I}} = \frac{g_2}{g_1} e^{-(E_2-E_1)/kT} = 5.06\times 10^{-9} \simeq \frac{1}{198{,}000{,}000}.$$

The result is that only one of every 200 million hydrogen atoms is in the first excited state and capable of producing Balmer absorption lines:

$$\frac{N_2}{N_{\mathrm{total}}} = \left(\frac{N_2}{N_1+N_2}\right)\left(\frac{N_{\mathrm{I}}}{N_{\mathrm{total}}}\right) = 5.06\times 10^{-9}.$$

We now turn to the calcium atoms. The ionization energy χ_{I} of Ca I is 6.11 eV, about half of the 13.6 eV ionization energy of hydrogen. We will soon see, however, that this small difference has a great effect on the ionization state of the atoms. Note that the Saha equation is very sensitive to the ionization energy because χ/kT appears as an *exponent* and $kT \approx 0.5$ eV $\ll \chi$. Thus a difference of several electron volts in the ionization energy produces a change of many powers of e in the Saha equation.

Evaluating the partition functions Z_{I} and Z_{II} for calcium is a bit more complicated than for hydrogen, and the results have been tabulated elsewhere:[15] $Z_{\mathrm{I}} = 1.32$ and $Z_{\mathrm{II}} = 2.30$. Thus the ratio of ionized to un-ionized calcium is

$$\left[\frac{N_{\mathrm{II}}}{N_{\mathrm{I}}}\right]_{\mathrm{Ca}} = \frac{2kTZ_{\mathrm{II}}}{P_e Z_{\mathrm{I}}}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_{\mathrm{I}}/kT} = 918.$$

Practically all of the calcium atoms are in the form of Ca II; only one atom out of 900 remains neutral. Now we can use the Boltzmann equation to estimate how many of these calcium ions are in the ground state, capable of forming the Ca II H and K absorption lines. The next calculation will consider the K ($\lambda = 393.3$ nm) line; the results for the H ($\lambda = 396.8$ nm) line are similar. The first excited state of Ca II is $E_2 - E_1 = 3.12$ eV above the ground state. The degeneracies for these states are $g_1 = 2$ and $g_2 = 4$. Thus the ratio of the number of Ca II ions in the first excited state to those in the ground state is

$$\left[\frac{N_2}{N_1}\right]_{\mathrm{Ca\ II}} = \frac{g_2}{g_1} e^{-(E_2-E_1)/kT} = 3.79\times 10^{-3} = \frac{1}{264}.$$

Out of every 265 Ca II ions, all but one are in the ground state and are capable of producing the Ca II K line. This implies that nearly *all* of the calcium atoms in the Sun's photosphere

[15]The values of the partition functions used here are from Aller (1963); see also Cox (2000), page 32.

are singly ionized and in the ground state,[16] so that almost all of the calcium atoms are available for forming the H and K lines of calcium:

$$\begin{aligned}\left[\frac{N_1}{N_{\text{total}}}\right]_{\text{Ca II}} &\simeq \left[\frac{N_1}{N_1+N_2}\right]_{\text{Ca II}}\left[\frac{N_{\text{II}}}{N_{\text{total}}}\right]_{\text{Ca}} \\ &= \left(\frac{1}{1+[N_2/N_1]_{\text{Ca II}}}\right)\left(\frac{[N_{\text{II}}/N_{\text{I}}]_{\text{Ca}}}{1+[N_{\text{II}}/N_{\text{I}}]_{\text{Ca}}}\right) \\ &= \left(\frac{1}{1+3.79\times 10^{-3}}\right)\left(\frac{918}{1+918}\right) \\ &= 0.995.\end{aligned}$$

Now it becomes clear why the Ca II H and K lines are so much stronger in the Sun's spectrum than are the Balmer lines. There are 500,000 hydrogen atoms for every calcium atom in the solar photosphere, but only an extremely small fraction, 5.06×10^{-9}, of these hydrogen atoms are un-ionized and in the first excited state, capable of producing a Balmer line. Multiplying these two factors,

$$(500{,}000) \times (5.06 \times 10^{-9}) \approx 0.00253 = \frac{1}{395},$$

reveals that there are approximately 400 times more Ca II ions with electrons in the ground state (to produce the Ca II H and K lines) than there are neutral hydrogen atoms with electrons in the first excited state (to produce the Balmer lines). The strength of the H and K lines is *not* due to a greater abundance of calcium in the Sun. Rather, the strength of these Ca II lines reflects the sensitive temperature dependence of the atomic states of excitation and ionization.

Figure 8.11 shows how the strength of various spectral lines varies with spectral type and temperature. As the temperature changes, a smooth variation from one spectral type to the next occurs, indicating that there are only minor differences in the composition of stars, as inferred from their spectra. The first person to determine the composition of the stars and discover the dominant role of hydrogen in the universe was Cecilia Payne (1900–1979). Her 1925 Ph.D. thesis, in which she calculated the relative abundances of 18 elements in stellar atmospheres, is among the most brilliant ever done in astronomy. In Section 9.5, we will see just how the relative abundances of atoms and molecules in stellar atmospheres are measured.

8.2 ■ THE HERTZSPRUNG–RUSSELL DIAGRAM

Early in the twentieth century, as astronomers accumulated data for an increasingly large sample of stars, they became aware of the wide range of stellar luminosities and absolute magnitudes. The O stars at one end of the Harvard sequence tended to be both brighter and

[16]It is left as an exercise to show that only a very small fraction of calcium atoms are doubly ionized (Ca III).

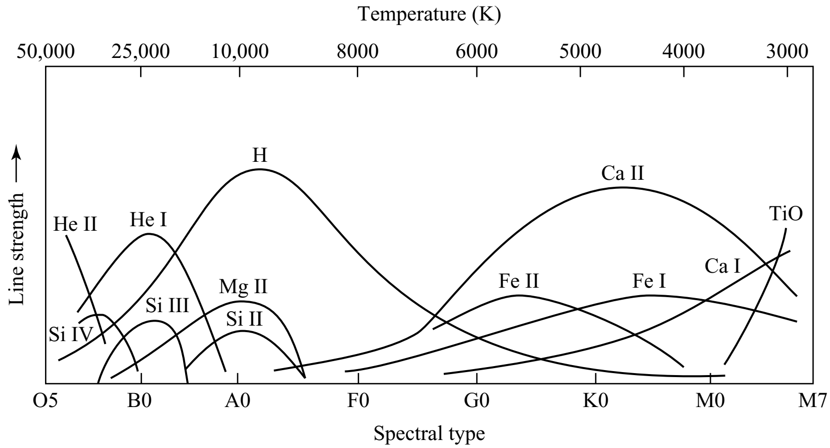

FIGURE 8.11 The dependence of spectral line strengths on temperature.

hotter than the M stars at the other end. In addition, the empirical mass–luminosity relation (see Fig. 7.7), deduced from the study of binary stars, showed that O stars are more massive than M stars. These regularities led to a theory of stellar evolution[17] that described how stars might cool off as they age. This theory (no longer accepted) held that stars begin their lives as young, hot, bright blue O stars. It was suggested that as they age, stars become less massive as they exhaust more and more of their "fuel" and that they then gradually become cooler and fainter until they fade away as old, dim red M stars. Although incorrect, a vestige of this idea remains in the terms *early* and *late* spectral types.

An Enormous Range in Stellar Radii

If this idea of stellar cooling were correct, then there should be a relation between a star's absolute magnitude and its spectral type. A Danish engineer and amateur astronomer, Ejnar Hertzsprung (1873–1967), analyzed stars whose absolute magnitudes and spectral types had been accurately determined. In 1905 he published a paper confirming the expected correlation between these quantities. However, he was puzzled by his discovery that stars of type G or later had a range of magnitudes, despite having the same spectral classification. Hertzsprung termed the brighter stars **giants**. This nomenclature was natural, since the Stefan–Boltzmann law (Eq. 3.17) shows that

$$R = \frac{1}{T_e^2}\sqrt{\frac{L}{4\pi\sigma}}. \tag{8.10}$$

If two stars have the same temperature (as inferred for stars having the same spectral type), then the more luminous star must be larger.

[17]Stellar evolution describes the change in the structure and composition of an individual star as it ages. This usage of the term *evolution* differs from that in biology, where it describes the changes that occur over generations, rather than during the lifetime of a single individual.

Hertzsprung presented his results in tabular form only. Meanwhile, at Princeton University, Henry Norris Russell (1877–1957) independently came to the same conclusions as Hertzsprung. Russell used the same term, *giant*, to describe the luminous stars of late spectral type and the term **dwarf** stars for their dim counterparts. In 1913 Russell published the diagram shown in Fig. 8.12. It records a star's observed properties: absolute magnitude on the vertical axis (with brightness increasing upward) and spectral type running horizontally (so temperature increases to the *left*). This first "Russell diagram" shows most of the features of its modern successor, the **Hertzsprung–Russell (H–R) diagram**.[18] More than 200 stars were plotted, most within a band reaching from the upper left-hand corner, home of the hot, bright O stars, to the lower right-hand corner, where the cool, dim M stars reside. This band, called the **main sequence**, contains between 80% and 90% of all stars in the H–R diagram. In the upper right-hand corner are the giant stars. A single **white dwarf**, 40 Eridani B, sits at the lower left.[19] The vertical bands of stars in Russell's diagram are a result of the discrete classification of spectral types. A more recent version of an observational H–R diagram is shown in Fig. 8.13 with the absolute visual magnitude of each star plotted versus its color index and spectral type.[20]

Figure 8.14 shows another version of the H–R diagram. Based on the average properties of main-sequence stars as listed in Appendix G, this diagram has a theorist's orientation: The luminosity and effective temperature are plotted for each star, rather than the observationally determined quantities of absolute magnitude and color index or spectral type. The uneven nature of the main sequence is an artifact of the slight differences among the references used to compile the tables in this appendix. The Sun (G2) is found on the main sequence, as is Vega (A0). Both axes are scaled logarithmically to accommodate the huge span of stellar luminosities, ranging from about $5 \times 10^{-4}\ \mathrm{L}_\odot$ to nearly $10^6\ \mathrm{L}_\odot$.[21] Actually, the main sequence is not a line but, rather, has a finite width, as shown in Figs. 8.12 and 8.13, owing to the changes in a star's temperature and luminosity that occur while it is on the main sequence and to slight differences in the compositions of stars. The giant stars occupy the region above the lower main sequence, with the **supergiants**, such as Betelgeuse, in the extreme upper right-hand corner. The white dwarfs (which, despite their name, are often not white at all) lie well below the main sequence.

The radius of a star can be easily determined from its position on the H–R diagram. The Stefan–Boltzmann law in the form of Eq. (8.10) shows that if two stars have the same surface temperature, but one star is 100 times more luminous than the other, then the

[18]The names of Hertzsprung and Russell were forever joined by another Danish astronomer, Bengt Strömgren (1908–1987), who suggested that the diagram be named after its two inventors. Strömgren's suggestion that star clusters be studied led to a clarification of the ideas of stellar evolution.

[19]Russell merely considered this star to be an extremely underluminous binary companion of the star 40 Eridani A; the extraordinary nature of white dwarfs was yet to be discovered. Note that the term *dwarf* refers to the stars on the main sequence and should not be confused with the *white dwarf* designation for stars lying well below the main sequence.

[20]Note that Fig. 8.13 suggests that a correlation exists between color index and spectral type, both of which are reflections of the effective temperature of the star. Recall that color index is closely related to the blackbody spectrum of a star (Section 3.6).

[21]Extremely late and early spectral types are not included in Fig. 8.14. The dimmest main-sequence stars are difficult to find, and the brightest have very short lifetimes, making their detection unlikely. As a result, only a handful of stars belonging to these classifications are known—too few to establish their average properties.

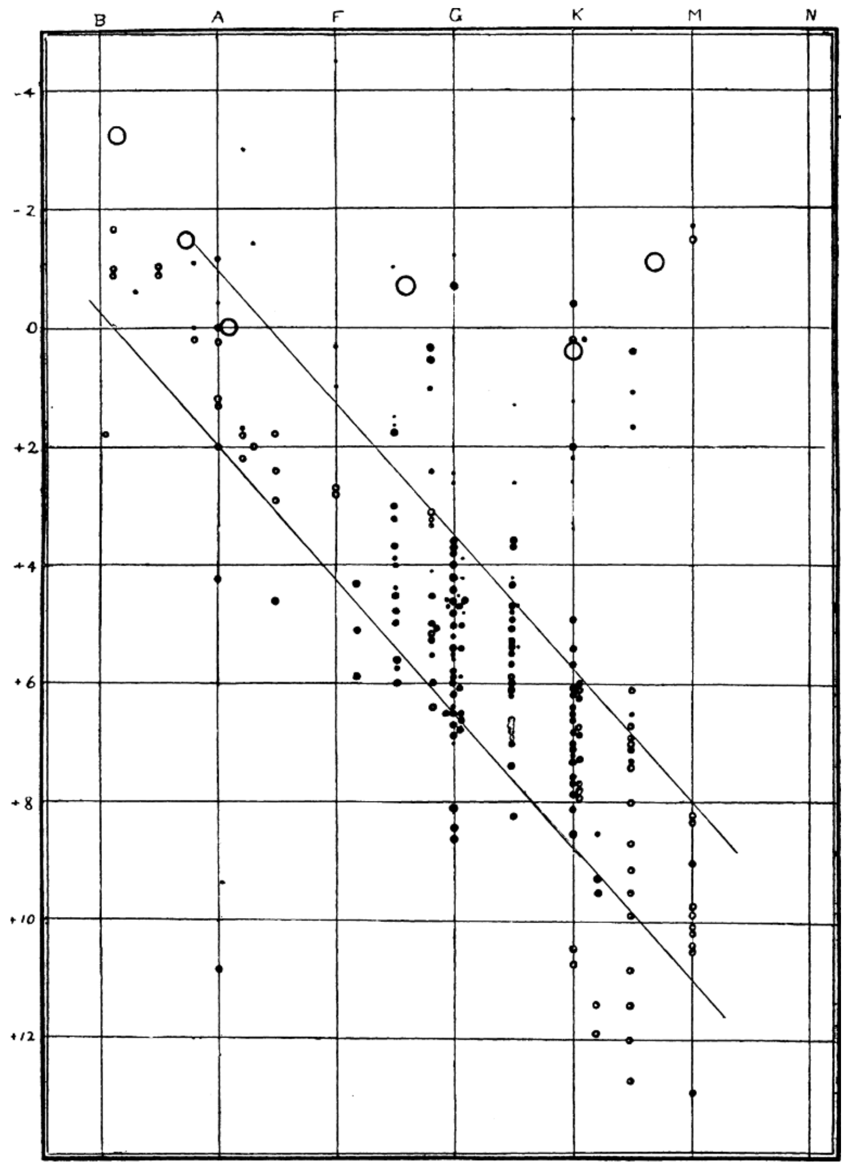

FIGURE 8.12 Henry Norris Russell's first diagram, with spectral types listed along the top and absolute magnitudes on the left-hand side. (Figure from Russell, *Nature*, *93*, 252, 1914.)

radius of the more luminous star is $\sqrt{100} = 10$ times larger. On a logarithmically plotted H–R diagram, the locations of stars having the same radii fall along diagonal lines that run roughly parallel to the main sequence (lines of constant radius are also shown in Fig. 8.14). The main-sequence stars show some variation in their sizes, ranging from roughly 20 $R_\odot$ at the extreme upper left end of the main sequence down to 0.1 $R_\odot$ at the lower right end. The giant stars fall between roughly 10 $R_\odot$ and 100 $R_\odot$. For example, Aldebaran (α Tauri),

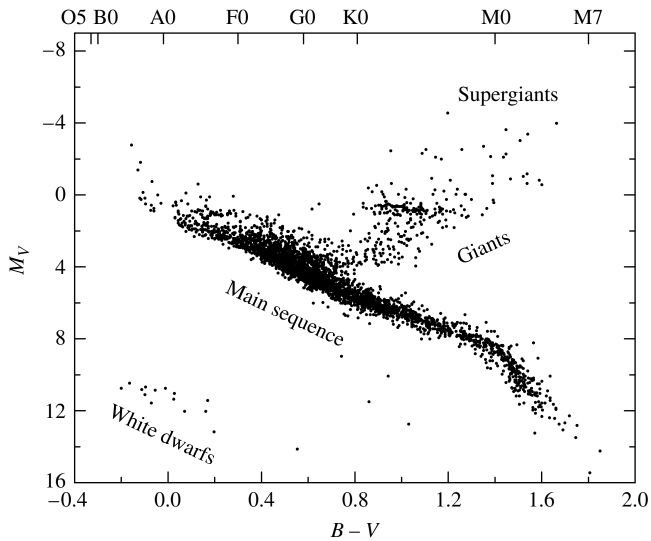

FIGURE 8.13 An observer's H–R diagram. The data are from the Hipparcos catalog. More than 3700 stars are included here with parallax measurements determined to better than 20%. (Data courtesy of the European Space Agency.)

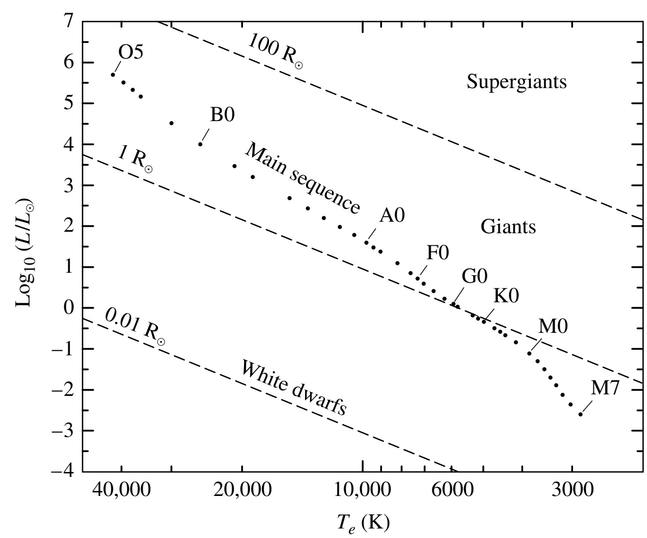

FIGURE 8.14 The theorist's Hertzsprung–Russell diagram. The dashed lines indicate lines of constant radius.

the gleaming "eye" of the constellation Taurus (the Bull), is an orange giant star that is 45 times larger than the Sun.

The supergiant stars are even larger. Betelgeuse, a pulsating variable star, contracts and expands roughly between 700 and 1000 times the radius of the Sun with a period of approximately 2070 days. If Betelgeuse were located at the Sun's position, its surface would at times extend past the orbit of Jupiter. The star μ Cephei in the constellation of Cepheus (a king of Ethiopia) is even larger and would swallow Saturn.[22]

The existence of such a simple relation between luminosity and temperature for main-sequence stars is a valuable clue that the position of a star on the main sequence is governed by a single factor. This factor is the star's *mass*.[23] The masses of stars along the main sequence are listed in Appendix G. The most massive O stars listed in that table are observed to have masses of 60 $M_\odot$,[24] and the lower end of the main sequence is bounded by M stars having at least 0.08 $M_\odot$.[25] Combining the radii and masses known for main-sequence stars, we can calculate the average density of the stars. The result, perhaps surprising, is that main-sequence stars have roughly the same density as water. Moving up the main sequence, we find that the larger, more massive, early-type stars have a *lower* average density.

Example 8.2.1. The Sun, a G2 main-sequence star, has a mass of $M_\odot = 1.9891 \times 10^{30}$ kg and a radius of $R_\odot = 6.95508 \times 10^8$ m. Its average density is thus

$$\overline{\rho}_\odot = \frac{M_\odot}{\frac{4}{3}\pi R_\odot^3} = 1410 \text{ kg m}^{-3}.$$

Sirius, the brightest-appearing star in the sky, is classified as an A1 main sequence star with a mass of 2.2 $M_\odot$ and a radius of 1.6 $R_\odot$. The average density of Sirius is

$$\overline{\rho} = \frac{2.2 \text{ M}_\odot}{\frac{4}{3}\pi (1.6 \text{ R}_\odot)^3} = 760 \text{ kg m}^{-3} = 0.54\,\overline{\rho}_\odot,$$

which is about 76 percent of the density of water. However, this is enormously dense compared to a giant or supergiant star. The mass of Betelgeuse is estimated to lie between 10 and 15 $M_\odot$; we will adopt 10 $M_\odot$ here. For illustration, if we take the maximum radius of this pulsating star to be about 1000 $R_\odot$, then the average density of Betelgeuse (at maximum size) is roughly

$$\overline{\rho} = \frac{10 \text{ M}_\odot}{\frac{4}{3}\pi (1000 \text{ R}_\odot)^3} = 10^{-8}\,\overline{\rho}_\odot!$$

Thus Betelgeuse is a tenuous, ghostly object—a hundred thousand times less dense than the air we breathe. It is difficult even to define what is meant by the "surface" of such a wraith-like star.

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[22] μ Cephei is a pulsating variable like Betelgeuse and has a period of 730 days. One of the reddest stars visible in the night sky, μ Cephei, is known as the *Garnet Star*.

[23] In Chapter 10 we will see how mass determines the location of a star on the main sequence.

[24] Theoretical calculations indicate that main-sequence stars as massive as 90 $M_\odot$ may exist, and recent observations have been made of a few stars with masses estimated near 100 $M_\odot$; see Section 15.3.

[25] Stars less massive than 0.08 $M_\odot$ have insufficient temperatures in their cores to support significant nuclear burning (see Chapter 10).

Morgan–Keenan Luminosity Classes

Hertzsprung wondered whether there might be some difference in the spectra of giant and main-sequence stars of the same spectral type (or same effective temperature). He found just such a variation in spectra among the stars cataloged by Antonia Maury. In her classification scheme she had noted line width variations that she referred to as a *c*-characteristic. The subtle differences in the relative strengths of spectral lines for stars of similar effective temperatures and different luminosities are depicted in Fig. 8.15. The work begun by Hertzsprung and Maury, and further developed by other astronomers, culminated in the 1943 publication of the *Atlas of Stellar Spectra* by William W. Morgan (1906–1994) and Phillip C. Keenan (1908–2000) of Yerkes Observatory. Their atlas consists of 55 prints of spectra that clearly display the effect of temperature and luminosity on stellar spectra and includes the criteria for the classification of each spectrum. The *MKK Atlas* established the *two-dimensional* Morgan–Keenan (M–K) system of spectral classification.[26] A **luminosity class**, designated by a Roman numeral, is appended to a star's Harvard spectral type. The numeral "I" (subdivided into classes Ia and Ib) is reserved for the supergiant stars, and "V" denotes a main-sequence star. The ratio of the strengths of two closely spaced lines is often employed to place a star in the appropriate luminosity class. In general, for stars of the same spectral type, narrower lines are usually produced by more luminous stars.[27] The Sun is a G2 V star, and Betelgeuse is classified as M2 Ia.[28] The series of Roman numerals extends below the main sequence; the subdwarfs (class VI or "sd") reside slightly to the left of the main sequence because they are deficient in metals. The M–K system does not extend to the white dwarfs, which are classified by the letter D. Figure 8.16 shows the corresponding divisions on the H–R diagram and the locations of a selection of specific stars, and Table 8.3 lists the luminosity classes.

The two-dimensional M–K classification scheme enables astronomers to locate a star's position on the Hertzsprung–Russell diagram based *entirely* on the appearance of its spectrum. Once the star's absolute magnitude, M, has been read from the vertical axis of the H–R diagram, the *distance* to the star can be calculated from its apparent magnitude, m, via Eq. (3.5),

$$d = 10^{(m-M+5)/5},$$

where d is in units of parsecs. This method of distance determination, called **spectroscopic parallax**, is responsible for many of the distances measured for stars,[29] but its accuracy is limited because there is not a perfect correlation between stellar absolute magnitudes and luminosity classes. The intrinsic scatter of roughly ± 1 magnitude for a specific luminosity class renders d uncertain by a factor of about $10^{1/5} = 1.6$.

[26]Edith Kellman of Yerkes printed the 55 spectra and was a co-author of the atlas; hence the additional "K" in *MKK Atlas*.

[27]In Section 9.5, we will find that because the atmospheres of more luminous stars are less dense, there are fewer collisions between atoms. Collisions can distort the energies of atomic orbitals, leading to broadening of the spectral lines.

[28]Betelgeuse, a pulsating variable star, is sometimes given the intermediate classification M2 Iab.

[29]Since the technique of parallax is not involved, the term *spectroscopic parallax* is a misnomer, although the name does at least imply a distance determination.

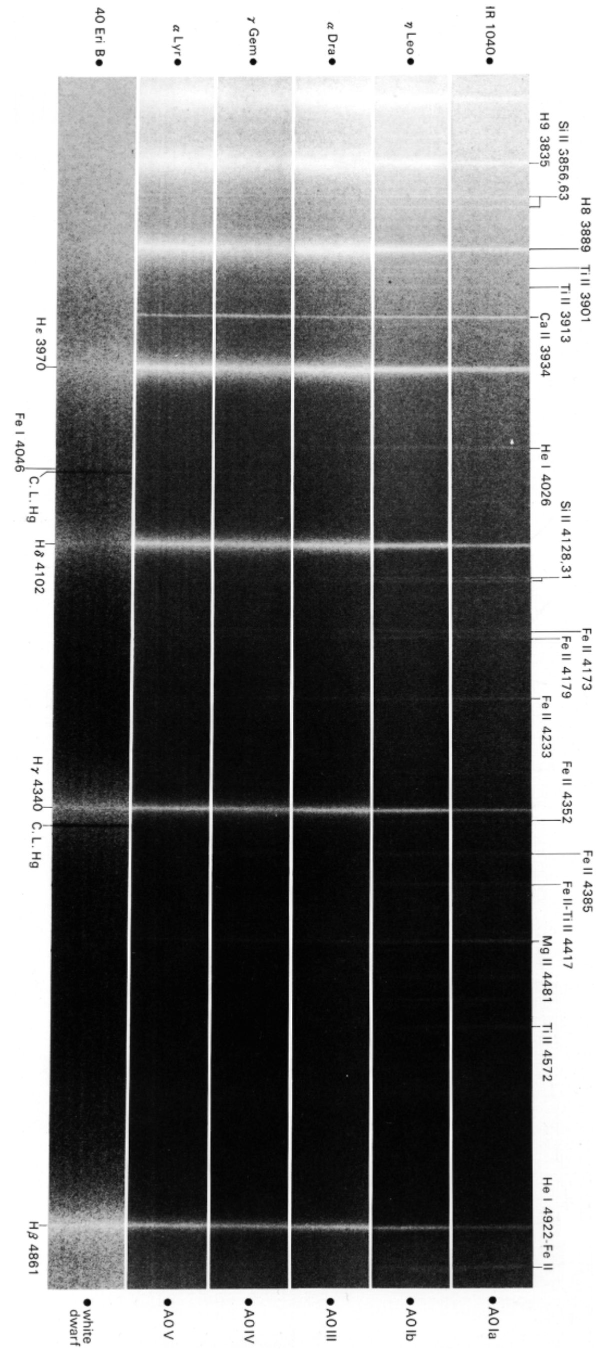

FIGURE 8.15 A comparison of the strengths of the hydrogen Balmer lines in types A0 Ia, A0 Ib, A0 III, A0 IV, A0 V, and a white dwarf, showing the narrower lines found in supergiants. These spectra are displayed as negatives, so absorption lines appear bright. (Figure from Yamashita, Nariai, and Norimoto, *An Atlas of Representative Stellar Spectra*, University of Tokyo Press, Tokyo, 1978.)

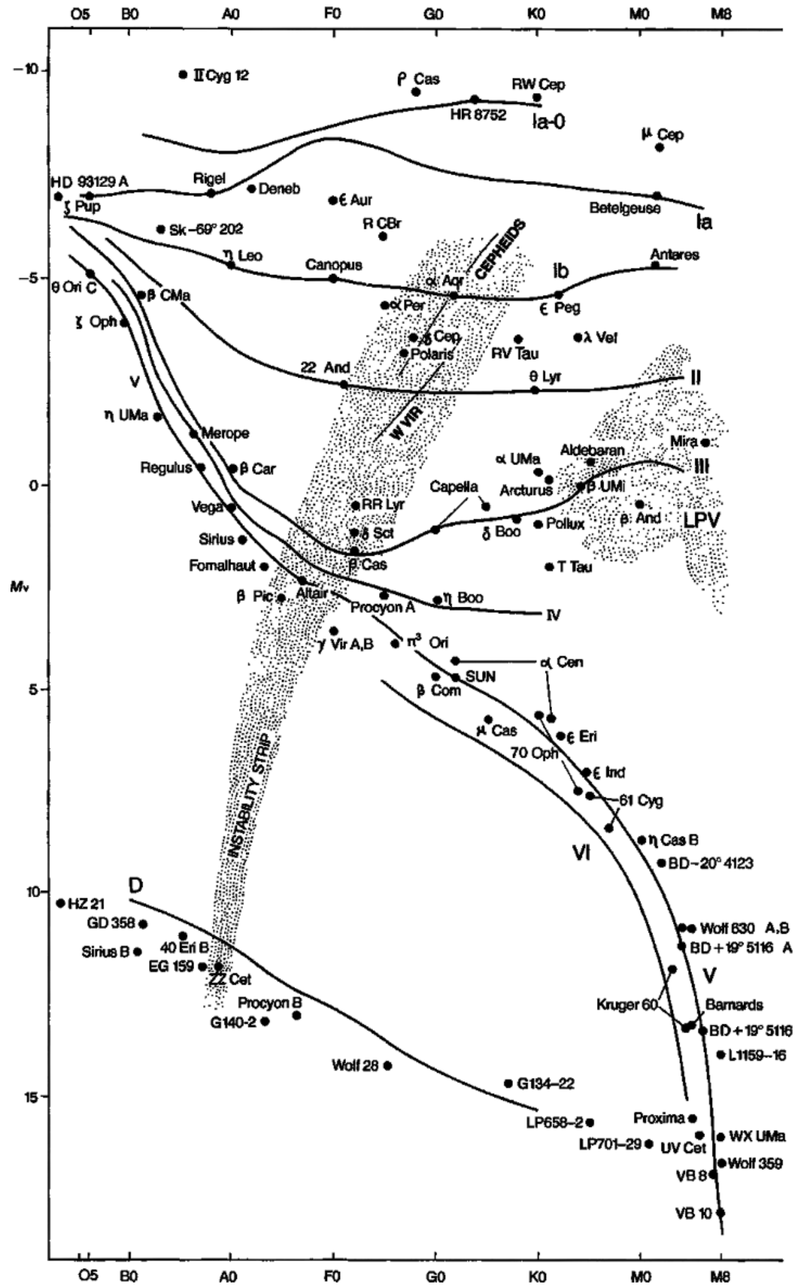

FIGURE 8.16 Luminosity classes on the H–R diagram. (Figure from Kaler, *Stars and Stellar Spectra*, © Cambridge University Press 1989. Reprinted with the permission of Cambridge University Press.)

TABLE 8.3 Morgan–Keenan Luminosity Classes.

Class	Type of Star
Ia-O	Extreme, luminous supergiants
Ia	Luminous supergiants
Ib	Less luminous supergiants
II	Bright giants
III	Normal giants
IV	Subgiants
V	Main-sequence (dwarf) stars
VI, sd	Subdwarfs
D	White dwarfs

SUGGESTED READING

General

Aller, Lawrence H., *Atoms, Stars, and Nebulae*, Third Edition, Cambridge University Press, New York, 1991.

Dobson, Andrea K., and Bracher, Katherine, "A Historical Introduction to Women in Astronomy," *Mercury*, January/February 1992.

Hearnshaw, J. B., *The Analysis of Starlight*, Cambridge University Press, Cambridge, 1986.

Herrmann, Dieter B., *The History of Astronomy from Hershel to Hertzsprung*, Cambridge University Press, Cambridge, 1984.

Hoffleit, Dorrit, "Reminiscenses on Antonia Maury and the *c*-Characteristic," *The MK Process at 50 Years*, Corbally, C. J., Gray, R. O., and Garrison, R. F. (editors), *ASP Conference Series*, *60*, 215, 1994.

Kaler, James B., *Stars and Their Spectra*, Cambridge University Press, Cambridge, 1997.

Technical

Aller, Lawrence H., *The Atmospheres of the Sun and Stars*, Ronald Press, New York, 1963.

Böhm-Vitense, Erika, *Stellar Astrophysics, Volume 2: Stellar Atmospheres*, Cambridge University Press, Cambridge, 1989.

Cox, Arthur N. (editor), *Allen's Astrophysical Quantities*, Fourth Edition, AIP Press, New York, 2000.

Geballe, T. R., et al., "Toward Spectral Classification of L and T Dwarfs: Infrared and Optical Spectroscopy and Analysis," *The Astrophysical Journal*, *564*, 466, 2002.

Kirkpatrick, J. Davy, et al., "Dwarfs Cooler Than "M": The Definition of Spectral Type "L" Using Discoveries From the 2-Micron All-Sky Survey (2MASS)," *The Astrophysical Journal*, *519*, 802, 1999.

Mihalas, Dimitri, *Stellar Atmospheres*, Second Edition, W.H. Freeman, San Francisco, 1978.

Novotny, Eva, *Introduction to Stellar Atmospheres and Interiors*, Oxford University Press, New York, 1973.

Padmanabhan, T., *Theoretical Astrophysics*, Cambridge University Press, Cambridge, 2000.

PROBLEMS

8.1 Show that at room temperature, the thermal energy $kT \approx 1/40$ eV. At what temperature is kT equal to 1 eV? to 13.6 eV?

8.2 Verify that Boltzmann's constant can be expressed in terms of electron volts rather than joules as $k = 8.6173423 \times 10^{-5}$ eV K^{-1} (see Appendix A).

8.3 Use Fig. 8.6, the graph of the Maxwell–Boltzmann distribution for hydrogen gas at 10,000 K, to estimate the fraction of hydrogen atoms with a speed within 1 km s^{-1} of the most probable speed, $v_{\rm mp}$.

8.4 Show that the most probable speed of the Maxwell–Boltzmann distribution of molecular speeds (Eq. 8.1) is given by Eq. (8.2).

8.5 For a gas of neutral hydrogen atoms, at what temperature is the number of atoms in the first excited state only 1% of the number of atoms in the ground state? At what temperature is the number of atoms in the first excited state 10% of the number of atoms in the ground state?

8.6 Consider a gas of neutral hydrogen atoms, as in Example 8.1.3.

(a) At what temperature will equal numbers of atoms have electrons in the ground state and in the second excited state ($n = 3$)?

(b) At a temperature of 85,400 K, when equal numbers (N) of atoms are in the ground state and in the first excited state, how many atoms are in the second excited state ($n = 3$)? Express your answer in terms of N.

(c) As the temperature $T \to \infty$, how will the electrons in the hydrogen atoms be distributed, according to the Boltzmann equation? That is, what will be the relative numbers of electrons in the $n = 1, 2, 3, \ldots$ orbitals? Will this in fact be the distribution that actually occurs? Why or why not?

8.7 In Example 8.1.4, the statement was made that "nearly all of the H I atoms are in the ground state, so Eq. (8.7) for the partition function simplifies to $Z_{\rm I} \simeq g_1 = 2(1)^2 = 2$." Verify that this statement is correct for a temperature of 10,000 K by evaluating the first three terms in Eq. (8.7) for the partition function.

8.8 Equation (8.7) for the partition function actually diverges as $n \to \infty$. Why can we ignore these large-n terms?

8.9 Consider a box of electrically neutral hydrogen gas that is maintained at a constant volume V. In this simple situation, the number of free electrons must equal the number of H II ions: $n_e V = N_{\rm II}$. Also, the total number of hydrogen atoms (both neutral and ionized), N_t, is related to the density of the gas by $N_t = \rho V/(m_p + m_e) \simeq \rho V/m_p$, where m_p is the mass of the proton. (The tiny mass of the electron may be safely ignored in this expression for N_t.) Let the density of the gas be 10^{-6} kg m^{-3}, typical of the photosphere of an A0 star.

(a) Make these substitutions into Eq. (8.8) to derive a quadratic equation for the fraction of ionized atoms:

$$\left(\frac{N_{\rm II}}{N_t}\right)^2 + \left(\frac{N_{\rm II}}{N_t}\right)\left(\frac{m_p}{\rho}\right)\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_{\rm I}/kT} - \left(\frac{m_p}{\rho}\right)\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_{\rm I}/kT} = 0.$$

(b) Solve the quadratic equation in part (a) for the fraction of ionized hydrogen, $N_{\rm II}/N_t$, for a range of temperatures between 5000 K and 25,000 K. Make a graph of your results, and compare it with Fig. 8.8.

8.10 In this problem, you will follow a procedure similar to that of Example 8.1.4 for the case of a stellar atmosphere composed of pure helium to find the temperature at the middle of the He I partial ionization zone, where half of the He I atoms have been ionized. (Such an atmosphere would be found on a white dwarf of spectral type DB; see Section 16.1.) The ionization energies of neutral helium and singly ionized helium are $\chi_{\rm I} = 24.6$ eV and $\chi_{\rm II} = 54.4$ eV, respectively. The partition functions are $Z_{\rm I} = 1$, $Z_{\rm II} = 2$, and $Z_{\rm III} = 1$ (as expected for any completely ionized atom). Use $P_e = 20$ N m^{-2} for the electron pressure.

(a) Use Eq. (8.9) to find $N_{\rm II}/N_{\rm I}$ and $N_{\rm III}/N_{\rm II}$ for temperatures of 5000 K, 15,000 K, and 25,000 K. How do they compare?

(b) Show that $N_{\rm II}/N_{\rm total} = N_{\rm II}/(N_{\rm I} + N_{\rm II} + N_{\rm III})$ can be expressed in terms of the ratios $N_{\rm II}/N_{\rm I}$ and $N_{\rm III}/N_{\rm II}$.

(c) Make a graph of $N_{\rm II}/N_{\rm total}$ similar to Fig. 8.8 for a range of temperatures from 5000 K to 25,000 K. What is the temperature at the middle of the He I partial ionization zone? Because the temperatures of the middle of the hydrogen and He I partial ionization zones are so similar, they are sometimes considered to be a single partial ionization zone with a characteristic temperature of 1–1.5 $\times 10^4$ K.

8.11 Follow the procedure of Problem 8.10 to find the temperature at the middle of the He II partial ionization zone, where half of the He II atoms have been ionized. This ionization zone is found at a greater depth in the star, and so the electron pressure is larger—use a value of $P_e = 1000$ N m^{-2}. Let your temperatures range from 10,000 K to 60,000 K. This particular ionization zone plays a crucial role in pulsating stars, as will be discussed in Section 14.2.

8.12 Use the Saha equation to determine the fraction of hydrogen atoms that are ionized, $N_{\rm II}/N_{\rm total}$, at the center of the Sun. Here the temperature is 15.7 million K and the number density of electrons is about $n_e = 6.1 \times 10^{31}$ m^{-3}. (Use $Z_{\rm I} = 2$.) Does your result agree with the fact that practically *all* of the Sun's hydrogen is ionized at the Sun's center? What is the reason for any discrepancy?

8.13 Use the information in Example 8.1.5 to calculate the ratio of doubly to singly ionized calcium atoms (Ca III/Ca II) in the Sun's photosphere. The ionization energy of Ca II is $\chi_{\rm II} = 11.9$ eV. Use $Z_{\rm III} = 1$ for the partition function of Ca III. Is your result consistent with the statement in Example 8.1.5 that in the solar photosphere, "nearly all of the calcium atoms are available for forming the H and K lines of calcium"?

8.14 Consider a giant star and a main-sequence star of the *same spectral type*. Appendix G shows that the giant star, which has a lower atmospheric density, has a slightly lower temperature than the main-sequence star. Use the Saha equation to explain why this is so. Note that this means that there is not a perfect correspondence between temperature and spectral type!

8.15 Figure 8.14 shows that a white dwarf star typically has a radius that is only 1% of the Sun's. Determine the average density of a 1-$M_\odot$ white dwarf.

8.16 The blue-white star Fomalhaut ("the fish's mouth" in Arabic) is in the southern constellation of Pisces Austrinus. Fomalhaut has an apparent visual magnitude of $V = 1.19$. Use the H–R diagram in Fig. 8.16 to determine the distance to this star.